

8

MEASUREMENT AND SPACE

FURTHER TRIGONOMETRY

Trigonometry is a branch of mathematics that uses the relationship between the angles and sides of triangles. Trigonometry is used in navigation when locating the position or bearing of a destination from a known location, and finding the distances between places that cannot be physically measured.

Trigonometry helps us understand physical phenomena that are periodic or cyclic, such as tidal movements, phases of the moon, average monthly temperatures and sound waves.

It is also used in construction, engineering, physics and gaming.



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Chapter outline

		Working mathematically				
8.01	Right-angled trigonometry	U	F	PS		
8.02	Bearings	U	F	PS	R	C
8.03	Pythagoras' theorem and trigonometry in 3D*	U	F	PS	R	C
8.04	Complementary relations and exact ratios*	U	F		R	
8.05	The trigonometric functions*	U	F		R	C
8.06	Trigonometric equations*	U	F			
8.07	The sine rule*	U	F	PS	R	
8.08	The sine rule for angles*	U	F	PS	R	C
8.09	The cosine rule*	U	F	PS	R	C
8.10	The cosine rule for angles*	U	F	PS	R	
8.11	The area of a triangle*	U	F	PS	R	
8.12	Problems involving the sine and cosine rules*	U	F	PS	R	C

***PATH**

Wordbank

bearing The angle used to show the direction of one location from a given point

complementary angles 2 angles that add to 90°

cosine rule A rule that relates the 3 sides and one of the angles of any triangle: $c^2 = a^2 + b^2 - 2ab\cos C$

exact ratio The sine, cosine and tangent of the special angles 30°, 45° and 60°, which can be expressed as exact fractions or surds rather than decimal approximations

included angle The angle between 2 known sides

quadrant A quarter of the number plane with the x- and y-axes as borders

sine rule A rule that relates the sides of any triangle to the sine of their opposite angles: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

unit circle A circle on the number plane with centre (0, 0) and radius 1

Quiz
Wordbank 8

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (25):

SkillCheck Trigonometry • Trigonometry on a calculator • Rounding angles

8.01 Finding an unknown side 1 • Finding the hypotenuse • Finding an unknown angle • Trigonometry • Trigonometry 2 • Angles of elevation and depression • Angle of depression 1 • Angle of depression 2

8.02 Trigonometry 2 • True bearings • Three-figure bearings • Bearings • Problems involving bearings

8.04 The exact trigonometric ratios

8.06 Trigonometric equations

8.07 The sine rule

8.08 Obtuse angles using the sine rule • The sine rule

8.09 The cosine rule

8.10 The cosine rule for angles 2 • The cosine rule • The cosine rule for angles 1

8.11 The sine area formula

8.12 The sine and cosine rules

Quizzes (5):

- Wordbank 8
- SkillCheck 8
- Mental skills 8
- Language of maths 8
- Test yourself 8

Worksheets (18):

SkillCheck Trigonometric calculations

8.01 Trigonometry review • Trigonometry problems

8.02 A page of bearings • NSW map bearings • Elevations and bearings • 16 points of the compass

8.04 The exact ratios

8.05 Unit circle investigation • The sine and cosine curves

8.06 Trigonometric equations

8.07 Discovering the sine rule

8.08 Sine rule problems

8.10 Cosine rule problems

8.12 Finding an unknown side 2 • Finding an unknown angle • The sine and cosine rules

Mind map Further trigonometry (Advanced)

Puzzles (11):

8.01 Trigonometry match-up • Trigonometry equations 1 • Trigonometry equations 2 • Finding an unknown angle • Trigonometry squaresaw • Solving triangles • Trigonometry: Finding angles

8.02 Bearings match-up

8.03 Pythagorean two-step problems

8.12 The sine and cosine rules

Language of maths Further trigonometry crossword (Advanced)

Technology (1):

8.05 Trigonometric graphs

Spreadsheet (1):

8.05 Trigonometric graphs

Presentations (2):

8.01 Applications of triangles

8.03 2D and 3D applications of trigonometry

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- use trigonometry to solve problems involving right-angled triangles, to find unknown sides and angles in right-angled triangles where the angle is measured in degrees and minutes
- solve trigonometry problems involving angles of elevation and depression, and bearings
- (PATH) use Pythagoras' theorem and trigonometry to solve three-dimensional problems
- (PATH) learn the complementary relations and the trigonometric ratios for 30°, 45°, 60°
- (PATH) use the unit circle to define and graph trigonometric functions
- (PATH) solve trigonometric equations
- (PATH) use the sine and cosine rules to find an unknown side or angle in any triangle
- (PATH) use the formula $A = \frac{1}{2}ab \sin C$ to find the area of a triangle

SkillCheck ANSWERS ON P. 652

1 Round each angle to the nearest degree.

a 64° 27′ **b** 25° 43′ **c** 12° 8′ 50″ **d** 85.85°

2 Evaluate each expression correct to 4 decimal places.

a cos 32° **b** sin 50.9° **c** tan 8° 45′ **d** 200 tan 18°

e 14 sin 87° 40′ **f** $\frac{13}{\cos 18° 27′}$ **g** $\frac{\sin 72° 43′}{28.4}$ **h** 5.9 tan 56.2°

3 Convert each angle to degrees and minutes, correct to the nearest minute.

a 45.80° **b** 33.1750° **c** 5.3460° **d** 84.375°

4 Find the size of angle A, correct to the nearest minute.

a $\cos A = \frac{3}{7}$ **b** $\tan A = 2.7$ **c** $\sin A = 0.4716$

Quiz
SkillCheck 8

Worksheet
Trigonometric calculations

Videos
Trigonometry
Trigonometry on a calculator
Rounding angles

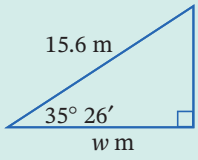
8.01

Right-angled trigonometry 8.01

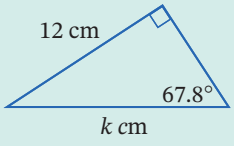
Example 1

Find the value of each variable, correct to one decimal place.

a



b



Solution

a $\cos 35° 26′ = \frac{w}{15.6}$

$w = 15.6 \cos 35° 26′$

$= 12.710...$

≈ 12.7 m

b $\sin 67.8° = \frac{12}{k}$

$k = \frac{12}{\sin 67.8°}$

$= 12.960...$

≈ 13.0 cm

Videos
Finding an unknown side 1
Finding the hypotenuse
Finding an unknown angle
Trigonometry

Puzzles
Trigonometry match-up
Trigonometry equations 1
Trigonometry equations 2
Finding an unknown angle
Trigonometry squaresaw
Solving triangles
Trigonometry: Finding angles

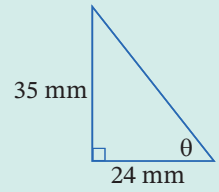
Worksheets
Trigonometry review
Trigonometry problems

Presentation
Applications of triangles

Video
Trigonometry 2

Example 2

Find the value of θ, correct to the nearest minute.



Solution

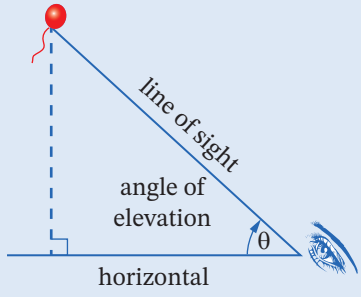
$\tan\theta = \frac{35}{24}$ — 35 is opposite and 24 mm is the adjacent side

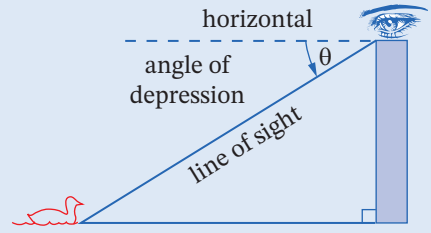
$\theta = 55.561...$ — SHIFT tan 35 [÷] 24 =

$= 55° 33′ 39.64″$ — Press °'" or DMS

$\approx 55° 34′$ — From the diagram, 55° 34′ seems reasonable

Angles of elevation and depression



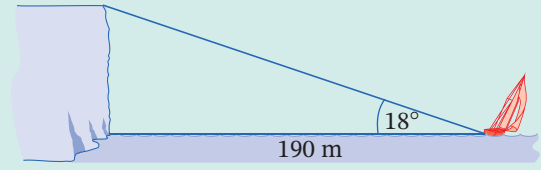


Problems involving angles of elevation and depression usually require the tan ratio in their solutions.

Video
Angles of elevation and depression

Example 3

The angle of elevation from a yacht to the top of a cliff is 18°. If the yacht is 190 m from the base of the cliff, find correct to one decimal place the height of the cliff.



Solution

Let the height be x metres.

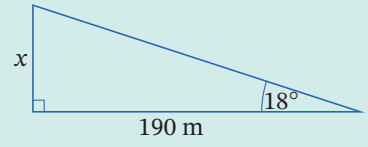
$\tan 18° = \frac{x}{190}$

$x = 190 \tan 18°$

$= 61.73474...$

≈ 61.7 m

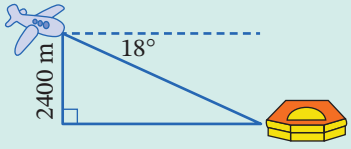
The height of the cliff is 61.7 m.



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Example 4

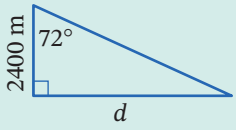
A pilot flying at a height of 2400 m saw a life raft in the sea at an angle of depression of 18°. Calculate the horizontal distance from the plane to the life raft, correct to the nearest metre.



Solution

Let d be the horizontal distance from the plane to the life raft.

The angle in the triangle adjacent to the angle of depression $= 90° - 18° = 72°$.

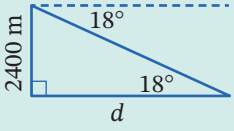


$$\tan 72° = \frac{d}{2400}$$
$$d = 2400 \tan 72°$$
$$= 7386.4404...$$
$$\approx 7386 \text{ m}$$

The horizontal distance from the plane to the life raft is 7386 m.

Alternative method

By alternate angles, the angle of elevation of the plane from the life raft is also 18°.



$$\tan 18° = \frac{2400}{d}$$
$$d = \frac{2400}{\tan 18°}$$
$$= 7386.4404...$$
$$\approx 7386 \text{ m}$$

Videos
Angles of elevation and depression
Angle of depression 1
Angle of depression 2

8.01

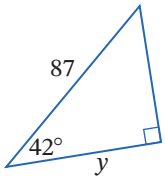
EXERCISE 8.01 ANSWERS ON P. 652

Right-angled trigonometry U F PS

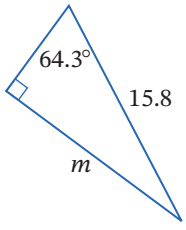
EXAMPLE 1

1 Calculate, correct to one decimal place, the value of each variable. All measurements are in centimetres.

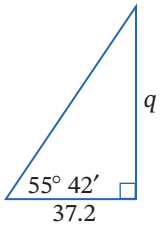
a



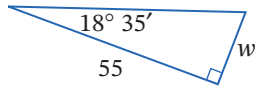
b



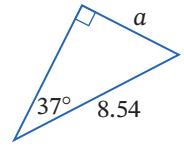
c



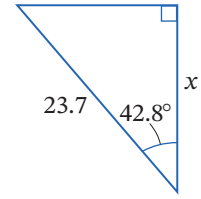
d



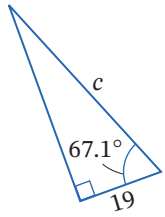
e



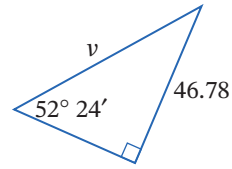
f



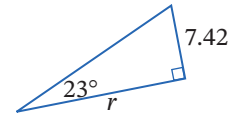
g



h



i



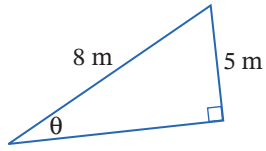
2 In $\triangle PQR$, $\angle P$ is a right angle, $\angle Q = 46° 12'$ and $PR = 8.7$ m. Find the length of QR.
Select the correct answer **A**, **B**, **C** or **D**.

A 6.3 m **B** 6.0 m **C** 12.1 m **D** 8.3 m

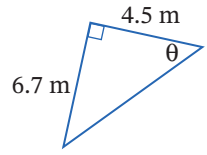
EXAMPLE 2

3 Find the value of θ, correct to the nearest minute.

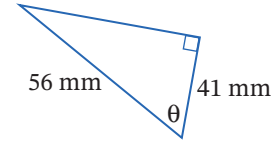
a



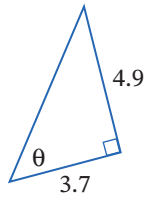
b



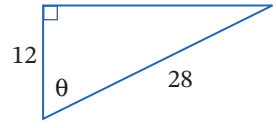
c



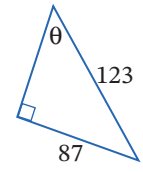
d



e



f



4 The entrance to a bank has a ramp for wheelchair access that is 3.6 m long. If the ramp is inclined at an angle of 10° to the ground, what is the height of the entrance (to the nearest cm)? PS

5 A section on a water slide has a vertical rise of 50 m and a horizontal run of 40 m. PS

a At what angle is the slide descending, correct to the nearest minute?

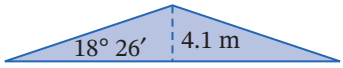
b What is the gradient of the section?

Foundation Standard Complex

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6 The training of a football team involves running up and down a sandhill 25 times. If the hill is 30 m high and inclined at an angle of 35° to the horizontal, how far does the team run during training? Answer correct to the nearest 0.1 km. PS

7 A roof has a pitch of 18° 26′. The height of the roof is 4.1 m. Calculate the overall width of the roof (correct to one decimal place). PS

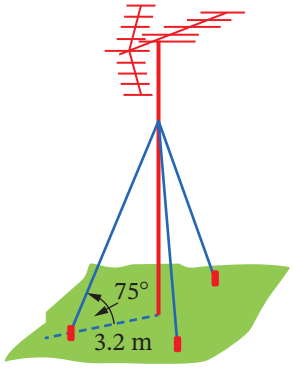


8 A glider is descending at an angle of 25° to the horizontal. The length of its flight path until it lands is 3.5 km. What was the altitude (to the nearest 0.1 km) of the glider? PS

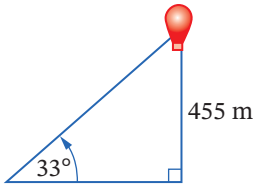
9 A radio tower is supported by cables as shown. The cables make an angle of 75° with the ground and are fixed 3.2 m from the base of the tower. Calculate, correct to one decimal place: PS

a the length of each cable

b how far up the tower each cable is fastened.



10 The angle of elevation of a hot air balloon at a height of 455 m is 33°. How far (to the nearest metre) has the balloon has drifted from its starting point? PS

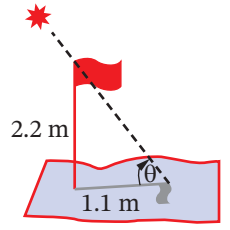


8.01

EXAMPLE 3

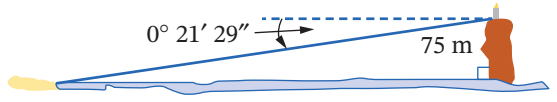
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11 A 2.2 m flag on a golf course casts a shadow 1.1 m long.
Find the angle of elevation of the sun (to the nearest degree). PS



EXAMPLE 4

12 The angle of depression of Collingwood Beach from Point Perpendicular is 0° 21′ 29″.
If the height of Point Perpendicular above sea level is 75 m, calculate the distance to the beach, correct to the nearest kilometre. PS



13 The angle of depression of a car from the top of a building is 41°.
If the car is 110 m from the base of the building, find the height of the building (correct to one decimal place). PS

14 From a lookout, the angle of depression to the centre of a town is 5° 47′. The horizontal distance from the centre of town to the lookout is 1.72 km. If the elevation (height above sea level) of the town is 690 m, calculate the elevation of the lookout. PS

15 Emily is walking towards a 72 m tall building. The angle of elevation to the top of the building is first 22° and 5 minutes later, 68°. How far has Emily walked towards the building? PS

16 From the window of Joshua's apartment, the angles of elevation and depression to the top and bottom of another apartment building are 42° and 32° respectively. If the buildings are 85 m apart, how tall, correct to the nearest metre, is the building that Joshua is looking at? PS

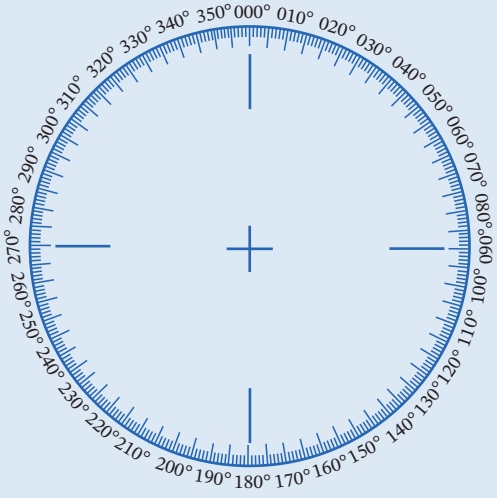
Bearings

8.02

Bearings are used in navigation. A **bearing** is an angle measurement used to precisely describe the direction of one location from a given reference point.

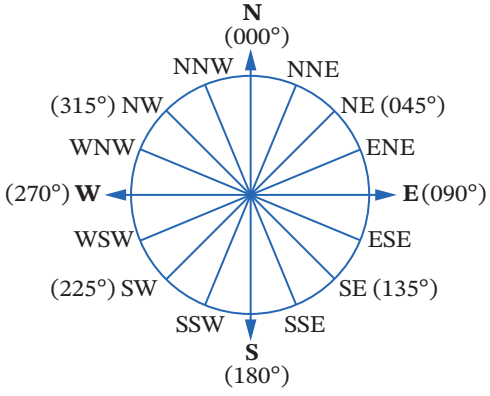
True bearings

True bearings, also called **three-figure bearings**, use angles from 000° to 360° to show the amount of turning measured **clockwise from north 000°**. Note that the angles are always written with 3 digits.



True bearings

This **compass rose** shows the true bearings of 16 points on the compass. A bearing of due east is 090°, while a compass direction of southwest (SW) is 225°.



N = north
E = east
S = south
W = west
NNE = north-northeast
ESE = east-southeast
SSW = south-southwest
WNW = west-northwest
NE = northeast
SE = southeast
SW = southwest
NW = northwest
ENE = east-northeast
SSE = south-southeast
WSW = west-southwest
NNW = north-northwest

Worksheets
A page of bearings
NSW map bearings
Elevations and bearings
16 points of the compass

Puzzle
Bearings match-up

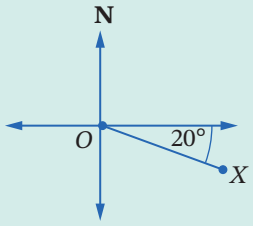
Videos
Trigonometry 2
True bearings

Video
Three-figure bearings

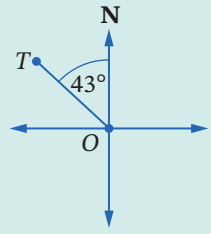
Example 5

Find the true bearing of each point from O.

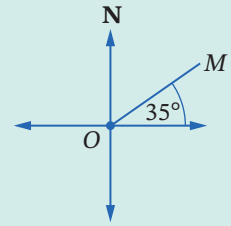
a



b



c



Solution

a Bearing of X from O is $90° + 20° = 110°$.

b Bearing of T from O is $360° - 43° = 317°$.

c Bearing of M from O is $90° - 35° = 055°$

This is sometimes written as 110°T to indicate that it is a True bearing.

Must be written as a 3-digit angle

Compass bearings

Compass bearings use acute angles measured from the North-South axis towards East or West, such as N20°E or S57°W. They are named according to the quadrant in which they are located, and are used in land surveying.

For the diagrams shown in Example 5 above:

- the compass bearing of X from O is S70°E
- the compass bearing of T from O is N43°W
- the compass bearing of M from O is N55°E

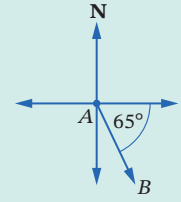
Video
Three-figure bearings

Example 6

Sketch point B on a compass rose if B has a bearing of 155° from A.

Solution

Draw the compass rose on the point where the bearing is measured from.
155° is between 90° and 180°, so B is in the southeast (SE) quadrant.
$155° - 90° = 65°$, so B is 65° from east (E).



Example 7

A plane leaves a town and remains on a bearing of 122° for 260 km.

a How far south of the town is the plane, correct to one decimal place?

b What is the bearing of the town from the plane?

Solution

a Let d km = distance south

$\angle SOP = 180° - 122°$ (angles on a straight line)

$= 58°$

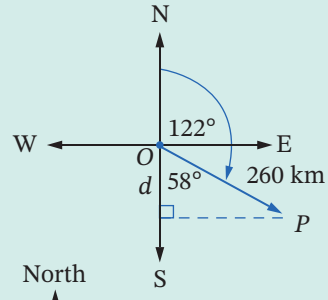
$\cos 58° = \frac{d}{260}$

$d = 260 \cos 58°$

$= 137.7790...$

≈ 137.8 km

The plane is 137.8 km south of the town.



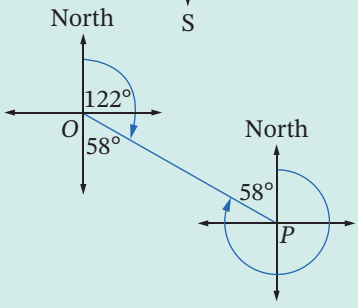
b Draw a compass rose with its centre at P.

$\angle OPN = 58°$ (alternate angles on parallel lines)

Bearing of O from $P = 360° - 58°$

$= 302°$

Bearing of the town from the plane = 302°



Videos
Bearings
Problems involving bearings

8.02

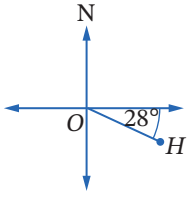
EXERCISE 8.02 ANSWERS ON P. 652

Bearings U F PS R C

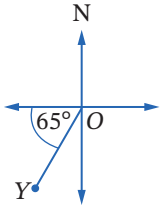
1 Write the true bearing of each point from O. c

EXAMPLE 5

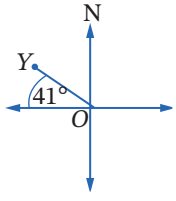
a



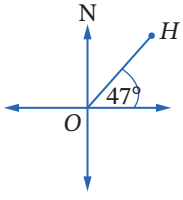
b



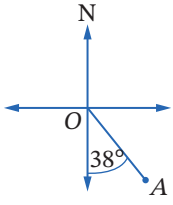
c



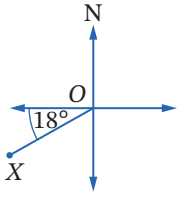
d



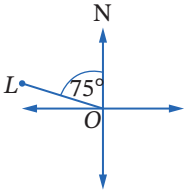
e



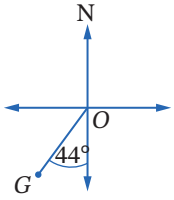
f



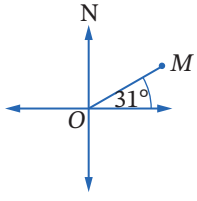
g



h



i



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2 Find the compass bearing of each point from O in Question **1**. C

3 **a** What is the compass direction halfway between east and southwest?

b What is the true bearing of this compass direction? C

4 Find the true bearing of each compass direction. C

a SW **b** E **c** NE **d** ESE

e WNW **f** W **g** SE **h** WSW

EXAMPLE 6

5 Sketch P on a compass rose if P has a bearing of: C

a 212° from G **b** 114° from X **c** 082° from A **d** 324° from W

6 If the bearing of A from B is 140°, what is the bearing of B from A? R

7 The bearing of H from P is 100°. What is the bearing of P from H? R

8 What is the (smallest) angle between:

a NW and SW? **b** NE and SSW **c** E and WSW **d** SW and WNW

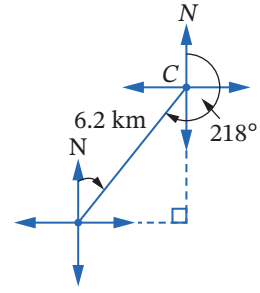
9 The compass bearing of H from M is N65°W. Find the compass bearing of M from H. R C

EXAMPLE 7

10 Annieke leaves a campsite (C) and rests after she has walked 6.2 km on a bearing of 218°. PS C

a How far south (correct to 2 decimal places) of the campsite did she stop to rest ?

b What is the bearing of the campsite from Annieke?



11 A yacht leaves Sydney and sails 780 km on a bearing of 071° to Lord Howe Island. PS C

a How far east (to the nearest km) of Sydney is Lord Howe Island?

b What is the bearing of Sydney from Lord Howe Island?

12 A plane flies from Sydney to Canberra on a bearing of S51.01°W. If the distance from Sydney to Canberra is 247 km, how far south is Canberra from Sydney? PS C

13 Klaas Creek is on a bearing of 314° from Hampervale and the distance between them is 158 km. Badgerton is 325 km from Hampervale and its bearing from Klaas Creek is 044°. Find the bearing of Badgerton from Hampervale. PS C

14 A plane flies 1200 km on a NE bearing and then changes direction to a bearing of NW and flies for a distance of 450 km. PS C

a How far, correct to the nearest km, is the plane from its starting point?

b What is the compass bearing (to the nearest minute) of the plane from its starting point?

15 A fishing boat leaves a harbour on a compass bearing of ESE and after 15 nautical miles, changes direction to a bearing of SSW and travels a further 10 nautical miles.

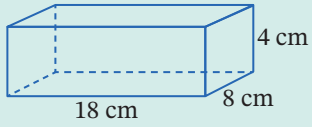
Find the true bearing of the harbour from the boat's final position, correct to the nearest minute. PS C

Pythagoras' theorem and trigonometry in 3D

8.03

Example 8

A wooden rectangular box has dimensions 18 cm × 8 cm × 4 cm.



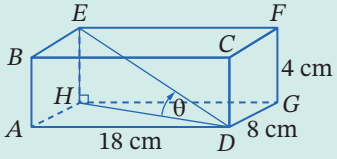
a Find, correct to one decimal place, the length of the longest pencil that can lie flat in the base of the box.

b Find, correct to one decimal place, the length of the longest diagonal of the box.

c Find, correct to the nearest degree, the angle that the longest diagonal makes with the base of the box.

Solution

Label the box as shown. HD is the length of the longest pencil that can lie flat in the base of the box, while ED is the longest diagonal of the box.



a Using the right-angled triangle DAH:

$HD^2 = DA^2 + AH^2$

$= 18^2 + 8^2$

$= 388$

$HD = \sqrt{388}$

$= 19.6977...$

≈ 19.7 cm

The longest pencil that can lie flat in the base of the box is 19.7 cm.

b $ED^2 = HD^2 + HE^2$

$ED^2 = \left(\sqrt{388}\right)^2 + 4^2$

$= 404$

$ED = \sqrt{404}$

$= 20.0997...$

≈ 20.1 cm

The longest diagonal of the box is 20.1 cm.

PATH

Puzzles
Pythagorean two-step problems

Presentation
2D and 3D applications of trigonometry

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c In the right-angled triangle EHD, θ is the angle that the longest diagonal makes with the base of the box.

$$\tan\theta = \frac{EH}{HD}$$
$$= \frac{4}{\sqrt{388}}$$
$$\theta = 11.4789...°$$
$$\approx 11°$$

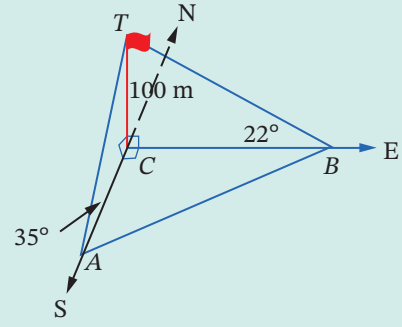
The longest diagonal makes an angle of 11° with the base of the box.

or use sin or cos since ED is also known

4 $\frac{\square}{\square}$ $\sqrt{\square}$ 388 SHIFT tan =

Example 9

A 100 m flagpole is observed from 2 different locations. From point A, due south of the flagpole the angle of elevation of the top of the flagpole is 35°; from point B, due east of the flagpole the angle of elevation is 22°. Find the distance between A and B, correct to the nearest metre.



Solution

There are 3 right-angled triangles in this diagram. To find AB, we must first find AC and CB using trigonometry.

In $\triangle ACT$,

$$\tan 35° = \frac{100}{AC}$$
$$AC = \frac{100}{\tan 35°}$$
$$= 142.8148...$$

Do not round yet.

In $\triangle BCT$,

$$\tan 22° = \frac{100}{CB}$$
$$CB = \frac{100}{\tan 22°}$$
$$\approx 247.5086...$$

In $\triangle ABC$,

$$AB^2 = AC^2 + CB^2$$
$$= (142.8148...)^2 + (247.5086...)^2$$
$$= 81656.6166...$$
$$AB = \sqrt{81656.6166...}$$
$$= 285.7562...$$
$$\approx 286 \text{ m}$$

From the diagram, 286 m seems to be a reasonable answer.

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EXERCISE 8.03 ANSWERS ON P. 653

PATH

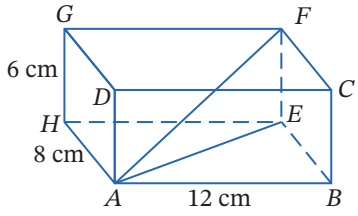
Pythagoras' theorem and trigonometry in 3D

U F PS R C

EXAMPLE 8

8.03

1 For this rectangular prism, calculate:

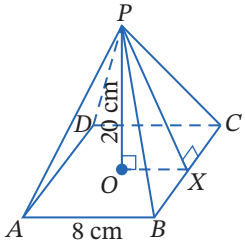


- **a** the length of AE in surd form
- **b** the length of AF, correct to one decimal place
- **c** the size of $\angle FAE$, correct to the nearest degree.

2 Sketch a diagram of a cube of length 20 cm, then find:

- **a** the exact length of the longest diagonal on any face
- **b** the length of the longest diagonal of the cube, correct to 2 decimal places
- **c** the angle that the longest diagonal makes with the base, correct to the nearest minute.

3 The diagram shows a square pyramid with base length 8 cm and perpendicular height 20 cm. PX is the slant height of the pyramid. Calculate: R



- **a** the length of PX, correct to 2 decimal places
- **b** the angle of inclination of PX, correct to the nearest degree.

4 A cone has a base diameter of 2.8 m and a slant height of 2.5 m. Find the angle that the cone makes with the vertical at the top of the cone. R

5 A fruit juice container is a square prism with dimensions 8 cm by 3 cm by 3 cm. R

- **a** Find, correct to one decimal place, the length of the longest drinking straw that fits inside the container.
- **b** To the nearest degree, what angle does the longest straw make with the vertical?

6 From a point X, 37 m from the base of a tree, the angle of elevation is 55° while the angle of elevation of the tree, from a point Y due east of the tree, is 25°. Find, correct to the nearest metre: PS R

- **a** the height of the tree
- **b** the distance of the tree from point Y

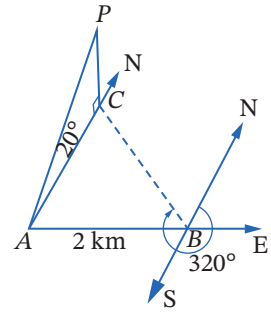
EXAMPLE 9

7 A 15 m flagpole stands on level ground. From point P, due west of the flagpole the angle of elevation of the top of the pole is 38°. From point Q, due north of the flagpole, the flagpole has an angle of elevation of 25°. Find PQ, correct to one decimal place. PS R

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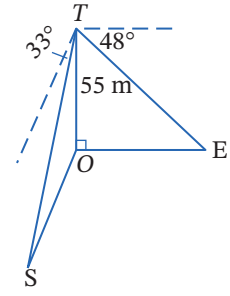
8 From a point, A, at the base of a mountain, the mountain peak, P, is due north and has an angle of elevation of 20°. From a point, B, 2 km due east of A on the same level, the mountain peak has a bearing of 320°. PS R



a What is the size of $\angle CBA$?

b Calculate the height, PC, of the mountain correct to the nearest metre.

9 From the top of her 55-metre office building, Madison observes 2 cars parked at ground level. The angle of depression of the car due east of the building is 48° and the angle of depression of the car parked due south of the building is 33°. PS R



Calculate, correct to the nearest metre, how far:

a each car is from the building

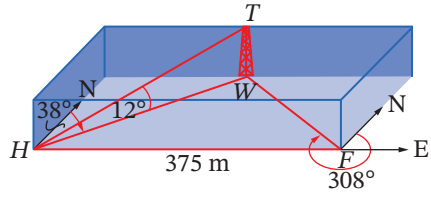
b the cars are apart.

Presentation
2D and 3D applications of trigonometry

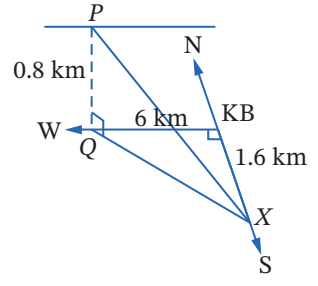
10 Hassan observes a transmission tower at an elevation angle of 12° and bearing 038°. Fatima stands 375 m due east of Hassan and observes the tower at a bearing of 308°. PS R

a Find the sizes of the angles of $\triangle FHW$.

b Find the height of the tower TW, correct to one decimal place.



11 A plane flies 6 km due west of Keira Bay at a constant height of 800 m. Xander sees the plane from his house 1.6 km south of Keira Bay.



Find, correct to the nearest degree: PS R C

a the true bearing of the plane from Xander

b the angle of elevation of the plane from Xander.

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12 Sophia observes a plane due north at an angle of elevation of 40°. The plane is flying due east at an altitude of 4000 m and at a speed of 200 km/h. PS R C

a Find the angle of elevation and the bearing of the plane from Sophia after 2 minutes.

b What is the distance (to the nearest m) from Sophia to the plane after 2 minutes?

13 From a lighthouse 95 m above sea level, a boat is observed on a bearing of SSE and at an angle of depression of 2°. Another boat is seen on a bearing of ENE and at angle of depression of 1.5°. PS R C

a Calculate the distance (correct to one decimal place) between the boats.

b What is the true bearing of the first boat from the second boat?

PATH

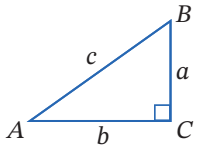
Complementary relations and exact ratios 8.04

Trigonometric ratios of complementary angles

PATH

In $\triangle ABC$, $\angle C = 90°$ so $\angle A + \angle B = 90°$ due to the angle sum of a triangle.

$\therefore A$ and B are complementary angles.



Now $\sin A = \frac{a}{c}$ and $\cos A = \frac{b}{c}$

But $\sin B = \frac{b}{c}$ and $\cos B = \frac{a}{c}$

$\therefore \sin A = \cos B$ and $\cos A = \sin B$

But $B = 90° - A$ (the complement of A)

$\therefore \sin A = \cos(90° - A)$ and $\cos A = \sin(90° - A)$

The sine and cosine ratios of complementary angles

$$\sin A = \cos(90° - A)$$
$$\cos A = \sin(90° - A)$$

Because of this complementary relationship, the sine and cosine ratios are called **complementary ratios**. That is why **co-sine** is named as the complement of **sine**.

PATH

Example 10

a If $\sin 35° = \cos \alpha$, find α.

b If $P + Q = 90°$ and $\sin Q = \frac{15}{17}$, find:

i $\cos P$ **ii** $\cos Q$

Solution

a $\sin 35° = \cos (90° - 35°)$ — $\sin A = \cos (90 - A)$

$= \cos 55°$

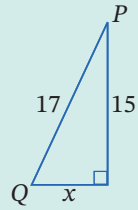
$\alpha = 55°$ — 35° and α are complementary angles

b i Since $P + Q = 90°$

$\cos P = \sin Q$ — P and Q are complementary angles

$= \frac{15}{17}$

ii Since $\sin Q = \frac{15}{17}$, draw a right-angled triangle to find $\cos Q$.



$x^2 = 17^2 - 15^2 = 64$

$x = \sqrt{64} = 8$

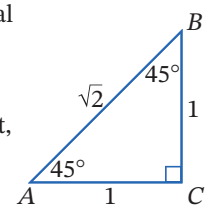
$\therefore \cos Q = \frac{8}{17}$

Exact trigonometric ratios of 30°, 45° and 60°



Worksheet
The exact ratios

The value of most trigonometric ratios can only be approximated in decimal form. However, the trigonometric ratios of the special angles 30°, 45° and 60° can be written in exact (surd) form. These are called the **exact ratios**.



For 45°, consider an isosceles right-angled triangle with 2 equal sides 1 unit, 2 equal angles 45° and hypotenuse h units.

By Pythagoras' theorem,

$h^2 = 1^2 + 1^2 = 2$

$h = \sqrt{2}$

$\therefore \sin 45° = \frac{1}{\sqrt{2}}, \cos 45° = \frac{1}{\sqrt{2}}, \tan 45° = \frac{1}{1} = 1$

For 30° and 60°, consider an equilateral triangle ABC with equal sides 2 units long, equal angles of 60° and an axis of symmetry CD. This means $\angle ACD = 30°$, $\angle ADC = 90°$ and side $AD = 1$ (half of 60°, 180° and 2 respectively).

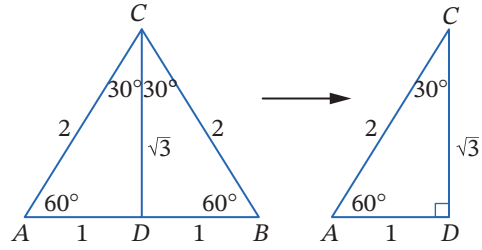
By Pythagoras' theorem,

$CD^2 = 2^2 - 1^2 = 3$

$CD = \sqrt{3}$

$\therefore \sin 30° = \frac{1}{2}, \cos 30° = \frac{\sqrt{3}}{2}, \tan 30° = \frac{1}{\sqrt{3}}$

$\sin 60° = \frac{\sqrt{3}}{2}, \cos 60° = \frac{1}{2}, \tan 60° = \sqrt{3}$



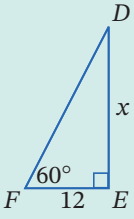
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The trigonometric ratios

	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

Note that $\sin 30° = \cos 60°$, $\sin 60° = \cos 30°$, and $\sin 45° = \cos 45°$, because 30° and 60° are complementary angles and 45° is the complement of itself.

Learn the values in this table by looking for patterns in the exact values.

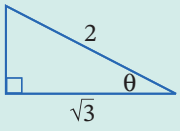


Example 11

Find the exact value of each variable.

a

b



Solution

a $\tan 60° = \frac{x}{12}$

$x = 12 \tan 60°$

$= 12\sqrt{3}$ $\quad \tan 60° = \sqrt{3}$

b $\cos \theta = \frac{\sqrt{3}}{2}$

$\theta = 30°$

EXERCISE 8.04 ANSWERS ON P. 653

Complementary relations and exact ratios U F R



1 Find the value of each variable.

a $\sin 47° = \cos X$ **b** $\cos 74° = \sin Y$ **c** $\sin 2.55° = \cos P$

d $\cos 55.2° = \sin \theta$ **e** $\sin 38° 17' = \cos M$ **f** $\cos 65° 26' = \sin T$

g $\sin (A + 28°) = \cos 37°$ **h** $\cos (M - 35°) = \sin 67°$ **i** $\sin 49° = \cos (T + 12°)$

2 Simplify each expression.

a $\cos 22° - \sin 68°$ **b** $\frac{\sin 57°}{\cos 33°}$

c $\frac{\sin (90° - \theta)}{\cos \theta}$ **d** $\frac{\cos 28° + \sin 62°}{\sin 62°}$

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PATH

3 If $\sin\alpha = \frac{5}{13}$ and $\alpha + \beta = 90°$, find $\cos\beta$, $\cos\alpha$ and $\sin\beta$. R

4 Given $\cos E = \frac{40}{41}$, and E and F are complementary angles, find $\sin F$, $\sin E$ and $\cos F$. R

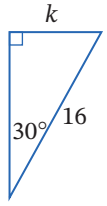
5 If $X + Y = 90°$, find the exact values of $\cos Y$, $\sin Y$ and $\sin X$, given $\cos X = \frac{2}{3}$. R

6 Find the exact values of $\cos\phi$, $\sin\phi$ and $\cos\theta$ if $\sin\theta = \frac{\sqrt{5}}{4}$ and $\theta = 90° - \phi$. R

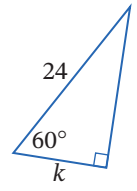
EXAMPLE 11

7 In each triangle, find the exact value of the variable.

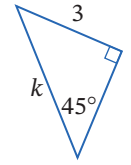
a



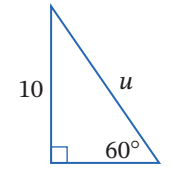
b



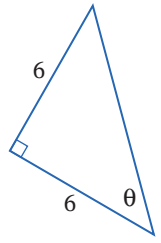
c



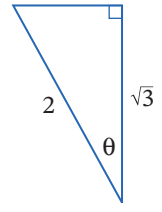
d



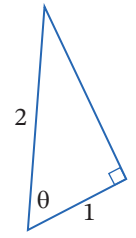
e



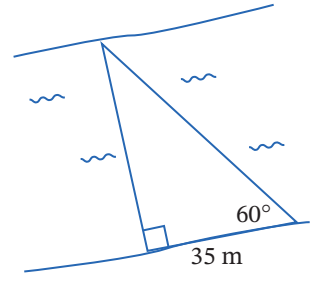
f



g

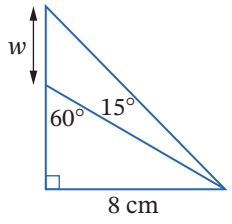


8 Find the width of this river in exact form.

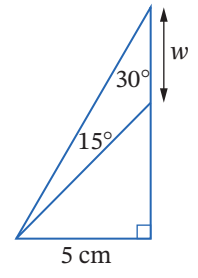


9 Find the exact value of w in each diagram.

a



b



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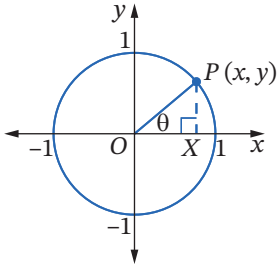
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The trigonometric functions 8.05

Trigonometric ratios of any angle

The sine, cosine and tangent ratios can be extended to include angles that are over 90°, that is, obtuse and reflex angles. The trigonometric ratios for angles of any size can be best explained using a **unit circle**.

A unit circle is a circle of radius 1 drawn on a number plane, with the origin as the centre of the circle. Starting from the positive direction of the x-axis, angles can be measured around this circle in an anti-clockwise direction.



Let $P(x, y)$ be any point on the unit circle as shown and θ the angle that PO makes with the positive x-axis.

Let the vertical interval from P meet the x-axis at X to make the right-angled triangle OXP.

Since P has coordinates (x, y), $OX = x$ and $XP = y$.

In $\triangle XOP$, $\cos\theta = \frac{OX}{OP} = \frac{x}{1}$ — $OP = 1$ because it is the radius of the unit circle

$\therefore \cos\theta = x$ — The x-coordinate of point P on the unit circle

Also, $\sin\theta = \frac{XP}{OP} = \frac{y}{1}$

$\therefore \sin\theta = y$ — The y-coordinate of the point P on a unit circle

and $\tan\theta = \frac{XP}{OX} = \frac{y}{x}$

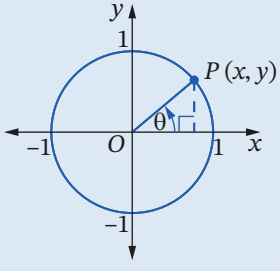
Trigonometric ratios on the unit circle

If $P(x, y)$ is any point on the unit circle, and θ is the angle that PO makes with the positive x-axis, then:

$\sin\theta = y$-coordinate of P

$\cos\theta = x$-coordinate of P

$\tan\theta = \frac{y\text{-coordinate of } P}{x\text{-coordinate of } P}$



PATH

8.05

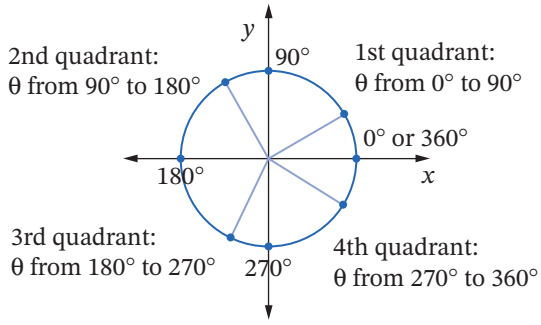
Worksheet
Unit circle investigation
The sine and cosine curves

Technology
Trigonometric graphs

Spreadsheet
Trigonometric graphs

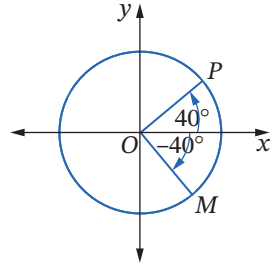
PATH

The x- and y-axes divide the number plane into 4 equal **quadrants**. Now we can investigate the trigonometric ratios for all angles from $\theta = 0°$ to $360°$, by looking at $P(x, y)$ on the unit circle in the 1st, 2nd, 3rd and 4th quadrants.



The unit circle can also be used to define the trigonometric ratios for angles below 0° and above 360°.

Negative angles (below 0°) are measured in a clockwise direction on the unit circle. In this diagram, M represents $-40°$ but it could also represent $360° - 40° = 320°$.



Angles above 360° are measured on the unit circle by going around the circle more than once. In the diagram, P represents 40°, but it could also represent $360° + 40° = 400°$.

The tangent ratio

The tangent ratio can be expressed in terms of the sine and cosine ratios.

$$\frac{\sin\theta}{\cos\theta} = \frac{y}{x}$$

But $\tan\theta = \frac{y}{x}$ $\quad\therefore \tan\theta = \frac{\sin\theta}{\cos\theta}$

The tangent ratio

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

Example 12

Given that $\sin\alpha = \frac{2}{\sqrt{13}}$ and $\cos\alpha = \frac{3}{\sqrt{13}}$, find $\tan\alpha$.

Solution

$$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$$
$$= \frac{2}{\sqrt{13}} \div \frac{3}{\sqrt{13}}$$
$$= \frac{2}{3}$$

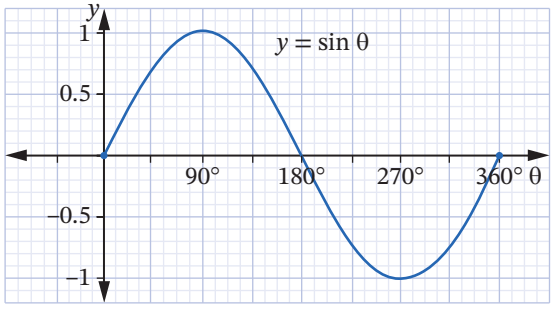
The sine curve

$\sin\theta$ = y-coordinate of P (the height of P above the x-axis), so note its value as θ moves around the unit circle (in an anti-clockwise direction) from 0° to 360°.

θ	0°	1st quadrant 0° to 90°	90°	2nd quadrant 90° to 180°	180°	3rd quadrant 180° to 270°	270°	4th quadrant 270° to 360°	360°
sin θ	0	from 0 to 1	1	from 1 to 0	0	from 0 to −1	−1	from −1 to 0	0

Note that the value of $\sin\theta$ always lies between 1 and −1.

The graph of $y = \sin\theta$ for θ from 0° to 360° is a 'wave curve' that repeats itself after 360°.



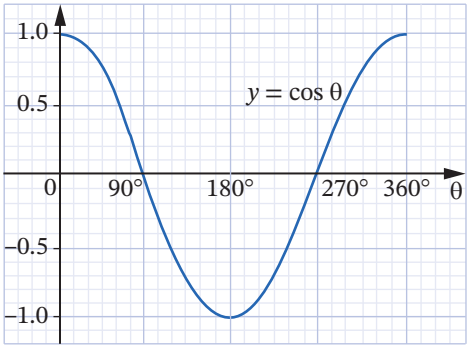
The cosine curve

$\cos\theta$ = x-coordinate of P (the length of P from the y-axis), so note its value for θ from 0° to 360°.

θ	0°	1st quadrant 0° to 90°	90°	2nd quadrant 90° to 180°	180°	3rd quadrant 180° to 270°	270°	4th quadrant 270° to 360°	360°
cos θ	1	from 1 to 0	0	from 0 to −1	−1	from −1 to 0	0	from 0 to 1	1

Note that the value of $\cos\theta$ always lies between 1 and −1.

The graph of $y = \cos\theta$ for θ from 0° to 360° is another 'wave curve' that repeats itself after 360°.



PATH

The tangent curve

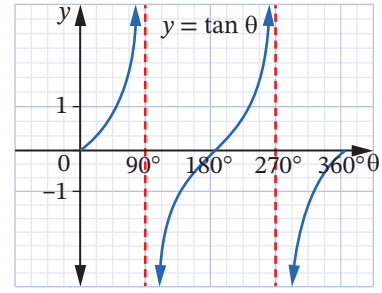
$\tan\theta = \frac{y\text{-coordinate of } P}{x\text{-coordinate of } P} = \frac{\sin\theta}{\cos\theta}$, so note its value for θ from 0° to 360°.

θ	0°	1st quadrant 0° to 90°	90°	2nd quadrant 90° to 180°	180°
tan θ	$\frac{0}{1} = 0$	$\frac{+}{+}$ = positive	$\frac{1}{0}$ = undefined	$\frac{+}{-}$ = negative	$\frac{0}{-1} = 0$

θ	3rd quadrant 180° to 270°	270°	4th quadrant 270° to 360°	360°
tan θ	$\frac{-}{-}$ = positive	$\frac{-1}{0}$ = undefined	$\frac{-}{+}$ = negative	$\frac{0}{1} = 0$

Note that the value of tan θ has no value at 90° and 270°.

The graph of $y = \tan\theta$ for θ from 0° to 360° is a curve that repeats itself after 180°, with asymptotes at θ = 90° and 270°.

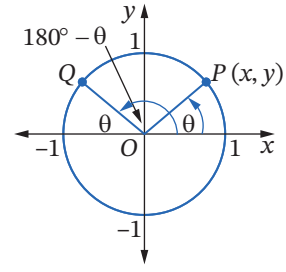


Trigonometric ratios of supplementary angles

For obtuse angles (between 90° and 180°, represented by the 2nd quadrant in the unit circle), we can use this diagram where Q is a reflection of P across the y-axis.

QO makes an angle of $180° - \theta$ with the positive x-axis.

In the 2nd quadrant, Q has a negative x-coordinate and a positive y-coordinate, so if the coordinates of P are (x, y), then the coordinates of Q are $(-x, y)$.



$\therefore \cos(180° - \theta) = -x$
$= -\cos\theta$

$\therefore \sin(180° - \theta) = y$
$= \sin\theta$

$\therefore \tan(180° - \theta) = \frac{y}{-x}$
$= -\frac{y}{x}$
$= -\tan\theta$

Supplementary relations

For obtuse angles (in the second quadrant), sine is positive while cosine and tangent are negative.

$$\sin(180° - A) = \sin A$$

The sine of an obtuse angle is equal to the **sine** of its supplement.

$$\cos(180° - A) = -\cos A$$

The cosine of an obtuse angle is equal to the **negative cosine** of its supplement.

$$\tan(180° - A) = -\tan A$$

The tangent of an obtuse angle is equal to the **negative tangent** of its supplement.

Example 13

If θ is acute, find θ if:

a $\tan 140° = -\tan \theta$ **b** $\sin 100° = \sin \theta$ **c** $\cos 120° = -\cos \theta$

Solution

a $\theta = 180° - 140°$
$= 40°$
$\therefore \tan 140° = -\tan 40°$

b $\theta = 180° - 100°$
$= 80°$
$\therefore \sin 100° = \sin 80°$

c $\theta = 180° - 120°$
$= 60°$
$\therefore \cos 120° = -\cos 60°$

Example 14

Find the exact value of each expression.

a $\sin 120°$ **b** $\tan 135°$ **c** $\cos 135°$

Solution

a $\sin 120° = \sin (180° - 120°)$
$= \sin 60°$
$= \frac{\sqrt{3}}{2}$

b $\tan 135° = -\tan (180° - 135°)$
$= -\tan 45°$
$= -1$

c $\cos 135° = \cos (180° - 135°)$
$= -\cos 45°$
$= -\frac{1}{\sqrt{2}}$

EXERCISE 8.05 ANSWERS ON P. 653

The trigonometric functions U F R C

EXAMPLE 12

1 a If $\sin A = \frac{60}{109}$ and $\cos A = \frac{91}{109}$, find $\tan A$

b If $\sin Y = 0.2$ and $\cos Y = 0.15$, find $\tan Y$

c If $\sin X = \frac{2}{\sqrt{13}}$ and $\cos X = \frac{3}{\sqrt{13}}$, find $\tan X$

d If $\cos P = \frac{40}{41}$ and $\sin P = \frac{9}{41}$, find $\tan P$

e If $\cos Q = \frac{3}{7}$ and $\sin Q = \frac{\sqrt{40}}{7}$, find $\tan Q$

f If $\cos X = \frac{60}{61}$ and $\tan X = \frac{11}{60}$, find $\sin X$

g If $\tan X = \frac{24}{7}$ and $\sin X = \frac{24}{25}$, find $\cos X$

h If $\tan X = \frac{2}{\sqrt{5}}$ and $\cos X = \frac{\sqrt{5}}{3}$, find $\sin X$

PATH

2 State whether each expression is positive (P) or negative (N).

a sin 95° **b** cos 46° **c** tan 153° **d** cos 171°

e sin 142° **f** tan 91° **g** tan 130° **h** cos 87°

3 Evaluate, correct to 2 decimal places, each trigonometric expression.

a cos 153° **b** tan 349° **c** sin 230°

d tan 173° 42′ **e** cos 300.9° **f** sin 324.8°

g sin 176° 54′ **h** cos 245° 23′ **i** tan (−38°)

j sin (−61°) **k** tan 370° **l** cos 434°

4 **a** Copy and complete this table of values for $y = \sin\theta$, evaluating y correct to 2 decimal places. R C

θ	0°	30°	60°	...	360°
y	0	0.5	0.87	...	0

b Graph $y = \sin\theta$, either by using graphing technology or on paper using a scale of 1 cm = 30° on the θ-axis and a scale of 4 cm = 1 unit on the y-axis.

c Comment on the shape of the graph $y = \sin\theta$. What are the maximum and minimum values of the graph and when do they occur?

d Does the graph have an axis of symmetry? If so, what is it?

e Does the graph have rotational symmetry? If so, what is the centre of symmetry?

f For what range of values of θ is sin θ:

i positive **ii** negative?

5 **a** Copy and complete this table of values for $y = \cos\theta$, evaluating y correct to 2 decimal places. R C

θ	0°	30°	60°	...	360°
y	1	0.87	0.5	...	1

b Graph $y = \cos\theta$, either by using graphing technology or on paper.

c Comment on the shape of the graph $y = \cos\theta$. What are the maximum and minimum values of the graph and when do they occur?

d Does the graph have an axis of symmetry? If so, what is it?

e Does the graph have rotational symmetry? If so, what is the centre of symmetry?

f For what range of values of θ is cos θ:

i positive **ii** negative?

g Comment on the similarities and differences between the graphs of $y = \sin x$ and $y = \cos x$.

EXAMPLE 13

6 If θ is acute, find θ if:

a $\cos 170° = -\cos\theta$ **b** $\sin 110° = \sin\theta$ **c** $\tan 130° = -\tan\theta$

d $\tan 97° = -\tan\theta$ **e** $\cos 115° = -\cos\theta$ **f** $\sin 168° = \sin\theta$

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7 Write each expression in terms of sin A, cos A or tan A, where A is acute.

a	cos 142°	**b**	sin 105°	**c**	cos 155°	**d**	tan 102°
e	cos 172.7°	**f**	sin 115.5°	**g**	cos 139° 35′	**h**	tan 170.8°
i	sin 120° 35′	**j**	tan 160° 10′	**k**	sin 95.5°	**l**	tan 139.5°

8 Find the exact value of each expression.

a	sin 150°	**b**	tan 135°	**c**	sin 135°	**d**	cos 120°
e	tan 150°	**f**	sin 120°	**g**	cos 150°	**h**	tan 120°
i	sin 90°	**j**	cos 135°				

9 **a** Copy and complete this table of values for $y = \tan\theta$, evaluating y correct to 2 decimal places. R C

θ	0°	30°	60°	...	360°
y	0	0.58	1.73	...	0

b Graph $y = \tan\theta$, either by using graphing technology or on paper.

c Comment on the shape of the graph $y = \tan\theta$. When does it start to repeat itself?

d Does the graph have an axis of symmetry? If so, what is it?

e Does the graph have rotational symmetry? If so, what is the centre of symmetry?

f For what range of values of θ is tan θ:

i positive **ii** negative?

PATH

EXAMPLE 14

8.06

Trigonometric equations 8.06

Example 15

Solve each trigonometric equation, giving all possible acute and obtuse solutions correct to the nearest degree.

a $\sin\theta = 0.7538$ **b** $\tan\theta = -2.5$

Solution

a $\sin\theta = 0.7538$

$\theta = 48.9206... \approx 49°$

On a calculator: SHIFT sin 0.7538 =

On some calculators, 2nd F instead of SHIFT

But θ could be obtuse, because sin θ is also positive in the second quadrant.

$\theta \approx 180° - 49° = 131°$

$\therefore \theta \approx 49°$ or 131°.

(Check: sin 49° = sin 131° = 0.7547...)

PATH

Video
Trigonometric equations

Worksheet
Trigonometric equations

Foundation Standard Complex

PATH

b $\tan\theta = -2.5$

$\theta = -68.1985... \approx -68°$

But θ is obtuse, because $\tan\theta$ is negative in the second quadrant.

$\theta \approx 180° - 68°$

$= 112°$

On a calculator: SHIFT tan (−) 2.5 =

On a calculator: 180 + Ans =

(Check: $\tan 112° = -2.4750...$)

Example 16

Solve each trigonometric equation correct to the nearest minute, if x is obtuse.

a $\cos x = -0.09$ **b** $\sin x = 0.64$

Solution

SHIFT cos automatically gives the obtuse angle when you enter a negative value.

a $\cos x = -0.09$

$x = 95.1636...$

$= 95° 9' 48.99''$

$\approx 95° 10'$

On a calculator: SHIFT cos (−) 0.09 =

On a calculator: °’” or DMS

b $\sin x = 0.64$

$x = 39.7918...$

But x is obtuse, so:

$x = 180 - 39.7918...$

$= 140.2081...$

$= 140° 12' 29.45''$

$\approx 140° 12'$

On a calculator: SHIFT sin 0.64 =

180 − Ans =

°’” or DMS

EXERCISE 8.06 ANSWERS ON P. 654

Trigonometric equations U F

EXAMPLE 15

1 Solve each trigonometric equation, giving all possible acute and obtuse solutions, correct to the nearest degree.

a $\sin\theta = 0.84$ **b** $\tan\theta = -\frac{3}{4}$ **c** $\cos\theta = -0.342$

d $\cos\theta = -\frac{7}{11}$ **e** $\sin\theta = 0.1164$ **f** $\tan\theta = -1$

g $\tan\theta = -5.8671$ **h** $\sin\theta = \frac{3}{7}$ **i** $\cos\theta = -0.4$

j $\sin\theta = \frac{3.8}{7}$ **k** $\cos\theta = -\frac{21}{80}$ **l** $\tan\theta = -\frac{15}{8}$

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2 Solve each trigonometric equation, correct to the nearest minute, if x is obtuse.
Note: some equations have no solution.

a $\sin x = \frac{4}{7}$ **b** $\sin x = 0.7438$ **c** $\sin x = 0.3514$

d $\sin x = 0.108$ **e** $\sin x = \frac{5}{11}$ **f** $\sin x = 0.9$

g $\cos x = 0.6$ **h** $\cos x = -0.6$ **i** $\tan x = 0.3$

j $\tan x = -0.3$ **k** $\sin x = 0.8$ **l** $\sin x = \frac{3}{7}$

3 Solve each trigonometric equation correct to the nearest degree, if x is between 0° and 180°.

a $\cos x = -\frac{8}{11}$ **b** $\tan x = -0.95$ **c** $\sin x = \frac{7}{8}$

d $4\cos x = \sqrt{2}$ **e** $3\sin x = 2$ **f** $-4\tan x = 3$

g $\tan x = 1$ **h** $\cos x = \frac{1}{2}$ **i** $\sin x = \frac{1}{\sqrt{2}}$

PATH

EXAMPLE 16

8.06

Mental skills 8: Maths without calculators ANSWERS ON P. 654

Quiz
Mental skills 8

Divisibility tests

A number is divisible by:	if:
2	its last digit is 2, 4, 6, 8 or 0
3	the sum of its digits is divisible by 3
4	its last two digits form a number divisible by 4
5	its last digit is 0 or 5
6	it is even and the sum of its digits is divisible by 3
9	the sum of its digits is divisible by 9
10	its last digit is 0

1 Test whether each number is divisible by 2, 3, 5 or 6.

a 250 **b** 189 **c** 78 **d** 465

e 1024 **f** 840 **g** 715 **h** 627

2 Test whether each number is divisible by 4, 9 or 10.

a 144 **b** 280 **c** 522 **d** 4170

e 936 **f** 726 **g** 342 **h** 5580

PATH

Investigation

Sides and opposite angles

This activity can also be done using dynamic geometry software.

1 Construct 3 triangles of different sizes.

2 Label the angles of each triangle A, B and C and the sides opposite them a, b and c respectively.

3 Measure the sides of each triangle correct to the nearest millimetre and the angles to the nearest degree. Copy the table below and record your results in it.

	Side a	Angle A	$\frac{a}{\text{Sin } A}$	Side b	Angle B	$\frac{b}{\text{Sin } B}$	Side c	Angle C	$\frac{c}{\text{Sin } C}$
Triangle 1									
Triangle 2									
Triangle 3									

4 Complete the table by calculating $\frac{a}{\sin A}$, $\frac{b}{\sin B}$ and $\frac{c}{\sin C}$ correct to 2 decimal places.

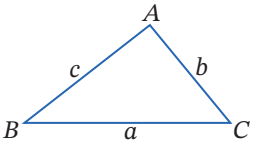
5 Compare the values of $\frac{a}{\sin A}$, $\frac{b}{\sin B}$ and $\frac{c}{\sin C}$ for each triangle. What do you notice?

8.07 The sine rule

PATH

Worksheet
Discovering the sine rule

The angles of a triangle are labelled with capital letters while the sides are labelled with lowercase letters. By convention, we use a to label the side opposite $\angle A$, b to label the side opposite $\angle B$, and so on.



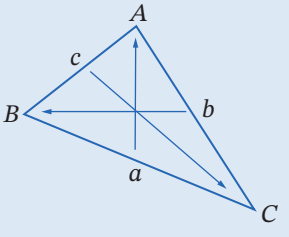
There is a relationship between each angle in a triangle and its opposite side. The longest side is always opposite the largest angle, the next smallest side is opposite the next smallest angle and so on. This relationship is called the **sine rule**.

The sine rule

For any triangle ABC:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

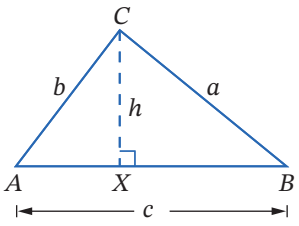
The ratios of the sides in a triangle to the sine of their opposite angles are equal.



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Proof:

In $\triangle ABC$, draw CX for the perpendicular height, h, of the triangle.
CX divides $\triangle ABC$ into 2 right-angled triangles.



In $\triangle AXC$, $\sin A = \frac{h}{b}$

$\therefore h = b \sin A$

In $\triangle BXC$, $\sin B = \frac{h}{a}$

$\therefore h = a \sin B$

$\therefore a \sin B = b \sin A$

$\frac{a \sin B}{\sin B} = \frac{b \sin A}{\sin B}$ Dividing both sides by $\sin B$

$a = \frac{b \sin A}{\sin B}$

$\frac{a}{\sin A} = \frac{b \sin A}{\sin B \sin A}$ Dividing both sides by $\sin A$

$\frac{a}{\sin A} = \frac{b}{\sin B}$

By drawing the perpendicular from A to BC, it can be shown that $\frac{b}{\sin B} = \frac{c}{\sin C}$

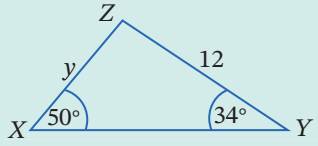
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

The sine rule allows us to apply trigonometry to any triangle, not just right-angled triangles.

The sine rule is used in problems involving 2 sides of a triangle and the 2 angles opposite them.

Example 17

Find y, correct to one decimal place.



Solution

$\frac{a}{\sin A} = \frac{b}{\sin B}$ Sides and opposite angles

$\frac{y}{\sin 34°} = \frac{12}{\sin 50°}$

$y = \frac{12 \sin 34°}{\sin 50°}$

$= 8.7596\ldots$

≈ 8.8 cm From the diagram, an answer of 8.8 cm looks reasonable

Video
The sine rule

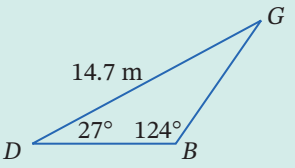
PATH

Example 18

In $\triangle BDG$, $\angle B = 124°$, $\angle D = 27°$ and $DG = 14.7$ m. Find BD, correct to one decimal place.

Solution

Draw a diagram.



$$\frac{BD}{\sin G} = \frac{14.7}{\sin 124°}$$

$\angle G = 180° - 27° - 124°$ (angle sum of a triangle)
$= 29°$

$$\frac{BD}{\sin 29°} = \frac{14.7}{\sin 124°}$$

$$BD = \frac{14.7 \sin 29°}{\sin 124°}$$
$= 8.5963...$
≈ 8.6 m

EXERCISE 8.07 ANSWERS ON P. 654

The sine rule U F PS R

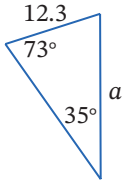
EXAMPLE 17

1 Evaluate each expression, correct to one decimal place.

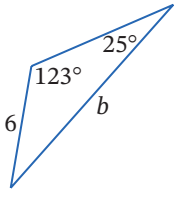
a $\frac{14.7 \sin 64°}{\sin 46°}$ **b** $\frac{34.5 \sin 33.4°}{\sin 115.7°}$ **c** $\frac{69 \sin 107° 33'}{\sin 38° 47'}$

2 Find the value of each variable, correct to 2 decimal places.

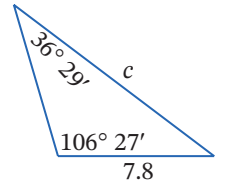
a



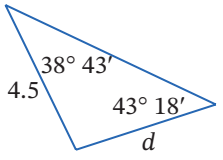
b



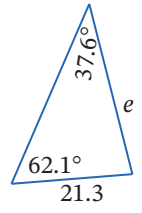
c



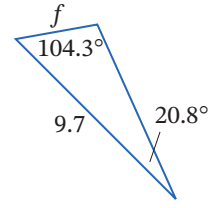
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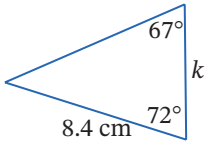
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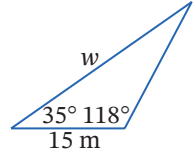
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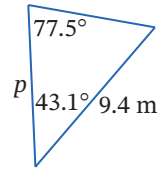
g



h



i



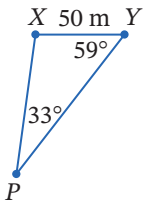
Foundation Standard Complex

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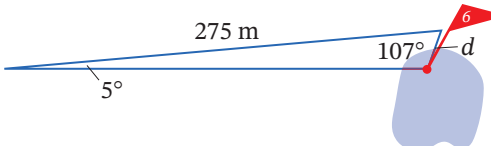
3 In $\triangle PTM$, $\angle P = 67°$, $\angle M = 52°$ and $PM = 12.5$ m. Find the length of PT.
Select the closest answer **A**, **B**, **C** or **D**.

A 10.7 m **B** 14.6 m **C** 11.3 m **D** 13.9 m

4 X and Y are 2 light towers 50 m apart on one side of a park. P is a light tower on the other side of the park. If $\angle Y = 59°$ and $\angle P = 33°$, find PX correct to the nearest metre. PS R



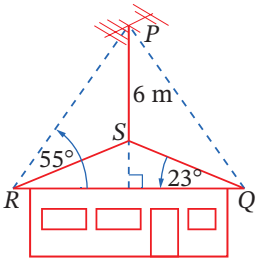
5 A golfer drives a ball 275 m at an angle of 5° off centre. The ball lands at an angle of 107° from the hole. Calculate the distance of the ball from the hole, correct to the nearest metre. PS R



6 A 6 m television antenna is mounted on a roof pitched at an angle of 23°. It is supported by 2 wires, PQ and PR, inclined at 55° to the horizontal. PS R

a Show that $\angle PSR = 113°$.

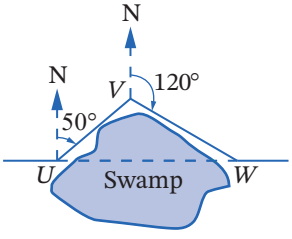
b Calculate the length of the wire PR, correct to the nearest centimetre.



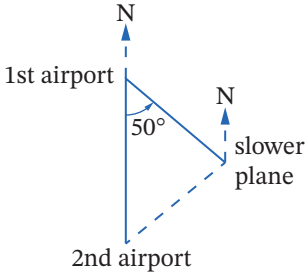
7 To avoid a swamp, Jesinta runs 70 m on a bearing of 050° to V. She then turns and runs to W on a bearing of 120°. If W is directly east of U: PS R

a find $\angle UVW$

b calculate UW, correct to one decimal place.

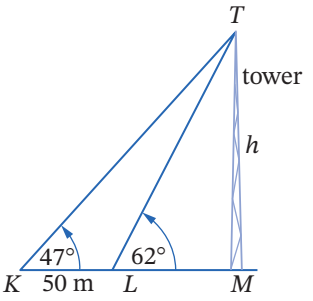


8 2 planes leave the airport at the same time. One flies due south at 400 km/h and lands at a second airport after $1\frac{1}{2}$ hours. The other flies on a bearing of S50°E and after $1\frac{1}{2}$ hours is at a bearing of N75°E from the second airport. How far (to the nearest km) is the slower plane from the second airport? PS R



PATH

9 The angle of elevation of a tower from a point L is 62°. From a point K, 50 m further from the tower, the angle of elevation is 47°. PS R



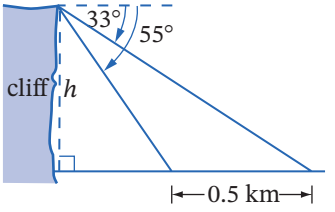
a Use the sine rule in $\triangle KTL$ to show that $TL = \dfrac{50\sin 47°}{\sin 15°}$.

b Let the height of the tower be h. In the right-angled $\triangle LMT$, show that $TL = \dfrac{h}{\sin 62°}$.

c Hence show that $h = \dfrac{50\sin 47°\sin 62°}{\sin 15°}$

d Hence calculate the height of the tower, correct to one decimal place.

10 From the top of a cliff, the angles of depression of 2 boats at sea that are 0.5 km apart are 55° and 33°. PS R



a Let the height of the cliff be h. Show that $h = \dfrac{0.5\sin 33°\sin 55°}{\sin 22°}$.

b Hence calculate the height, to the nearest metre.

11 A yacht leaves Fremantle and sails on a bearing of 312° for 650 km. It then changes direction and sails on a bearing of 235° until it is due west of Fremantle. How far is the yacht from Fremantle (to the nearest km)? PS R



Hilke Maunder/Alamy Stock Photo

Foundation Standard Complex

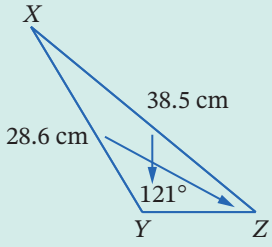
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The sine rule for angles

8.08

Example 19

Find angle Z, correct to the nearest degree.



Solution

$$\frac{28.6}{\sin Z} = \frac{38.5}{\sin 121°}$$

Sides and opposite angles

$$\frac{\sin Z}{28.6} = \frac{\sin 121°}{38.5}$$

Inverting both sides so that Z is in the numerator

$$\sin Z = \frac{28.6 \sin 121°}{38.5} = 0.636\ldots$$

On a calculator: SHIFT sin Ans =

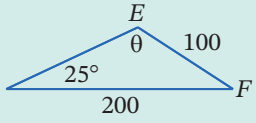
$$Z = 39.55...$$

$$\approx 40°$$

From the diagram, an answer of 40° looks reasonable

Example 20

Find θ correct to the nearest minute if it is an obtuse angle.



Solution

$$\frac{200}{\sin\theta} = \frac{100}{\sin 25°}$$

$$\frac{\sin\theta}{200} = \frac{\sin 25°}{100}$$

$$\sin\theta = \frac{200 \sin 25°}{100}$$

$$= 0.845\ldots$$

$$\theta = 57.697...$$

But θ is obtuse, so:

$$\theta = 180 - 57.697...$$

$$= 122.3027...$$

$$\approx 122° 18'$$

PATH

Worksheet
Sine rule problems

Video
Obtuse angles using the sine rule

8.08

PATH

The ambiguous case (when there are 2 possible answers)

When we use the sine rule to find an angle, it is possible to find both an acute angle and an obtuse angle as solutions. Likewise, there could be 2 possible triangles: one acute-angled, the other obtuse-angled. However, the obtuse-angled triangle may not be possible. We need to check that the sum of the angles in the triangle is not greater than 180°.

Video
The sine rule

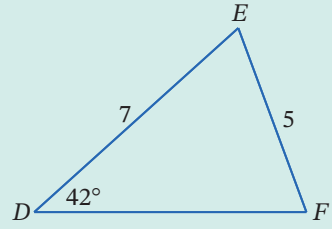
Example 21

a In $\triangle DEF$, $\angle D = 42°$, $d = 5$ cm and $f = 7$ cm. Find $\angle F$ correct to the nearest degree.

b In $\triangle LMN$, $\angle M = 130°$, $LN = 15$ cm and $LM = 7$ cm.
Find $\angle N$, correct to the nearest degree.

Solution

a Draw a rough diagram.



$$\frac{7}{\sin F} = \frac{5}{\sin 42°}$$

$$\frac{\sin F}{7} = \frac{\sin 42°}{5}$$

$$\sin F = \frac{7 \sin 42°}{5} = 0.93678\ldots$$

$$F = 69.5181\ldots \approx 70°$$

But F could be obtuse.

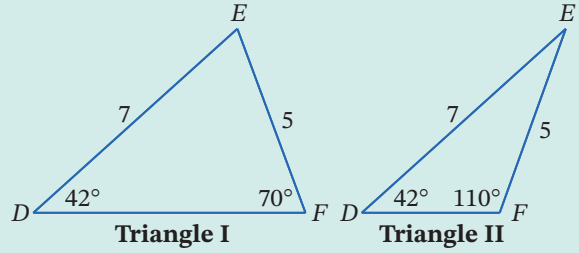
$$F = 180° - 70° = 110°$$

Checking the third angle of the obtuse-angled triangle:

$\angle E = 180° - 42° - 110° = 28°$

$\therefore$ The obtuse-angled solution is possible.

$\therefore \angle F = 70°$ or $110°$



b

$$\frac{\sin N}{7} = \frac{\sin 130°}{15}$$

$$\sin N = \frac{7 \sin 130°}{15} = 0.3574\ldots$$

$$N = 20.9459\ldots \approx 21°$$

But N could be obtuse.

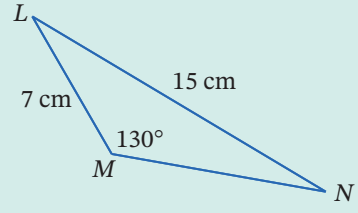
$N = 180° - 21° = 159°$

Checking the third angle of the obtuse-angled triangle:

$\angle L = 180° - 130° - 159° = -109°$

$\therefore$ The obtuse-angled solution is not possible.

$\therefore \angle N = 21°$



Impossible

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EXERCISE 8.08 ANSWERS ON P. 654

The sine rule for angles U F PS R C

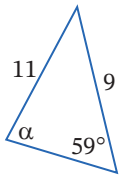
1 Find the acute angle X in each equation, correct to the nearest degree.

a $\sin X = \frac{5.3 \sin 123°}{9.7}$ **b** $\sin X = \frac{39 \sin 85°29'}{64}$ **c** $\sin X = \frac{467 \sin 63.8°}{518}$

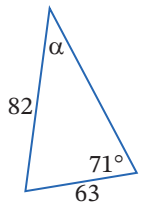
2 Find α in each triangle if α is acute, correct to the nearest 0.1 degree.

EXAMPLE 19

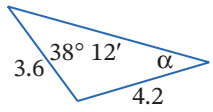
a



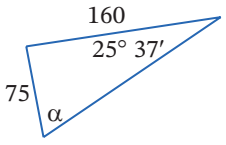
b



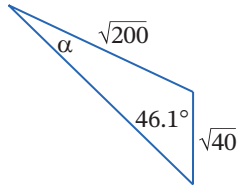
c



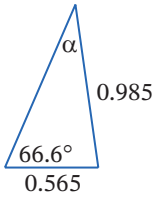
d



e



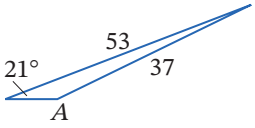
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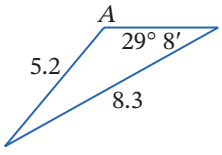
3 Find the size of $\angle A$ to the nearest minute if $\angle A$ is obtuse.

EXAMPLE 20

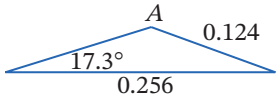
a



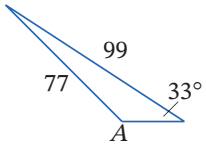
b



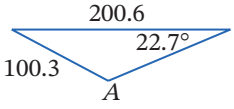
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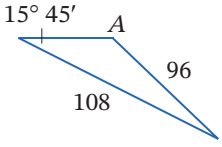
d



e



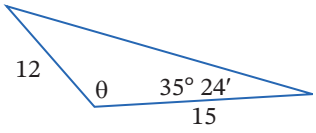
f



4 Find the size of θ. Select the correct answer **A**, **B**, **C** or **D**.

A 46° 24′ **B** 98° 12′

C 133° 36′ **D** 81° 48′



5 In $\triangle BGP$, $\angle G = 121°$, $BG = 5.2$ m, and $BP = 12.8$ m. Find $\angle B$ (to the nearest degree).

6 Find all possible angles for each triangle, correct to the nearest degree. R C

EXAMPLE 21

a In $\triangle PQR$, $\angle P = 35°$, $p = 8$ cm, and $q = 10$ cm. Find $\angle Q$.

b In $\triangle UVW$, $\angle W = 95°$, $w = 16$ cm, and $v = 10$ cm. Find $\angle V$.

c In $\triangle XYZ$, $\angle Y = 24°$, $y = 3.4$ km, and $z = 5.7$ km. Find $\angle Z$.

d In $\triangle DEF$, $\angle E = 37°$, $e = 107$ mm, and $d = 121$ mm. Find $\angle D$.

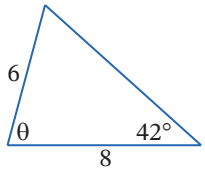
Foundation Standard Complex

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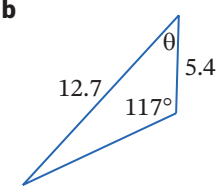
PATH

7 Find θ in each triangle correct to the nearest degree, given that θ is acute.

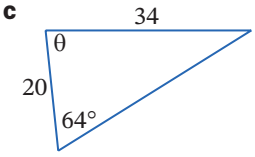
a



b



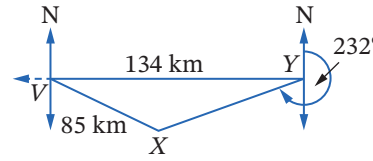
c



8 Max walked 5.6 km due east from his campsite. He then turned and walked on a bearing of S33°W until he was 6 km from the campsite. What is the bearing of the campsite from Max? Select **A**, **B**, **C** or **D**. PS R C

A N52°E **B** N71°W **C** N19°W **D** N52°W

9 Yorker (Y) is 134 km due east of Vendor (V). Xena (X) is on a bearing of 232° from Yorker and is also 85 km from Vendor. $\angle VXY$ is an obtuse angle. Find the bearing of Xena from Vendor. PS R C



8.09 The cosine rule

PATH

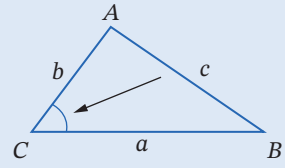
The **cosine rule** is a relationship between the 3 sides of a triangle and one of its angles. It is an extension of Pythagoras' triangle that can be applied to any triangle, not just right-angled ones.

The cosine rule

For any triangle ABC:

$$c^2 = a^2 + b^2 - 2ab\cos C$$

where c is the unknown side, C is the angle opposite c, and a and b are the other 2 sides.



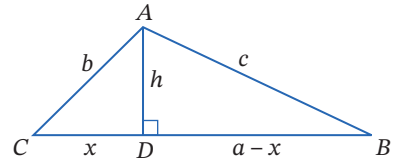
Proof:

In $\triangle ABC$, draw AD for the perpendicular height, h, of the triangle. AD divides $\triangle ABC$ into 2 right-angled triangles.

Let $CD = x$, $\therefore DB = a - x$

In $\triangle ADC$, $b^2 = h^2 + x^2$

$$\therefore h^2 = b^2 - x^2 \qquad [1]$$



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In $\triangle ABD$, $c^2 = h^2 + (a - x)^2$

$\therefore h^2 = c^2 - (a - x)^2$

$\therefore b^2 - x^2 = c^2 - (a - x)^2$

$b^2 - x^2 = c^2 - (a^2 - 2ax + x^2)$

$b^2 - x^2 = c^2 - a^2 + 2ax - x^2$

$b^2 = c^2 - a^2 + 2ax$

$\therefore c^2 = a^2 + b^2 - 2ax$ [2]

In $\triangle ADC$, $\cos C = \frac{x}{b}$

$\therefore x = b \cos C$

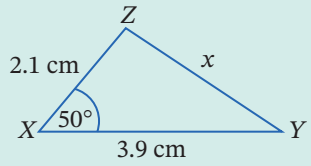
Substituting for x in [2]:

$c^2 = a^2 + b^2 - 2ab \cos C$

The cosine rule can be used in problems involving 3 sides of a triangle and one of the angles.

Example 22

Find x correct to 2 decimal places.



Solution

$c^2 = a^2 + b^2 - 2ab \cos C$

$x^2 = 2.1^2 + 3.9^2 - 2 \times 2.1 \times 3.9 \cos 50°$

$= 9.091138...$

$x = \sqrt{9.091138...}$

≈ 3.02 cm

50° is the angle opposite x

From the diagram, an answer of 3.02 cm looks reasonable

Video
The cosine rule

EXERCISE 8.09 ANSWERS ON P. 654

The cosine rule U F PS R C

1 Solve each equation for x, correct to one decimal place.

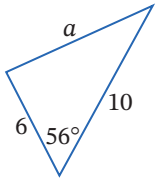
a $x^2 = 8^2 + 9^2 - 2 \times 8 \times 9 \times \cos 38°$

b $x^2 = 11.3^2 + 9.7^2 - 2 \times 11.3 \times 9.7 \times \cos 76.9$

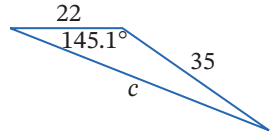
c $x^2 = 17^2 + 20.1^2 - 2 \times 17 \times 20.1 \times \cos 149° 45'$

2 Find, correct to 2 decimal places, the value of each variable.

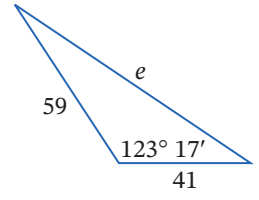
a



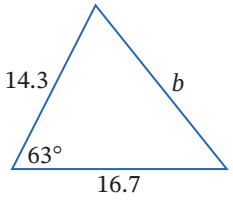
b



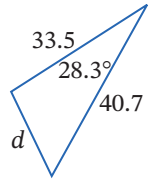
c



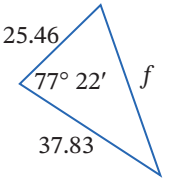
d



e



f



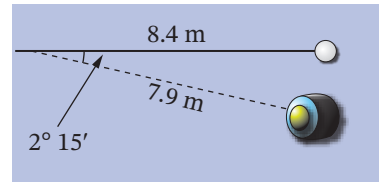
Foundation | Standard | Complex

PATH

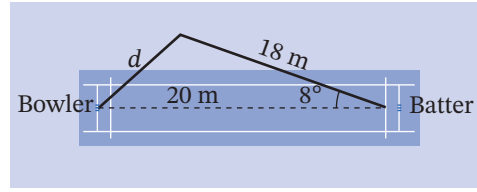
3 In $\triangle DEF$, $\angle E = 38° 46'$, $d = 8.7$ m and $f = 15.6$ m. What is the length of DF?
Select the closest answer **A**, **B**, **C** or **D**.

A 10.4 m **B** 19.5 m **C** 17.9 m **D** 12.8 m

4 In a game of lawn bowls, Jayden is aiming to hit the jack (target ball) 8.4 m away. If he bowls 2° 15′ off centre and his bowl travels 7.9 m, how far is his bowl from the jack? Answer correct to one decimal place. PS R

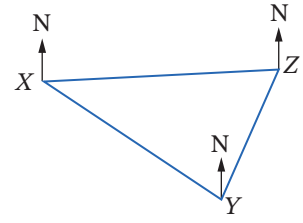


5 In a cricket match, the distance between the bowler and the batter was 20 m. During one bowl, the batter hit the ball at an angle of 8° to the line of the pitch and the bowler ran and caught the ball after it had travelled 18 m. How far did the bowler run to catch the ball?
Select **A**, **B**, **C** or **D**. PS R



A 1.1 m **B** 2.0 m **C** 3.3 m **D** 4.0 m

6 A yacht sails from X to Y on a bearing of 130° for 4.2 km. It then turns and travels to Z on a bearing of 025° for 2.9 km. PS R C

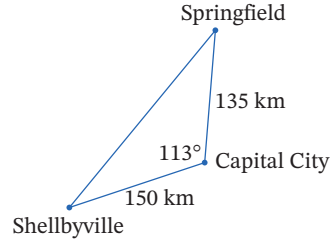


a Copy the diagram and mark the given information on it.

b Explain why $\angle XYZ = 75°$.

c Calculate the distance XZ, correct to one decimal place.

7 3 towns are joined by straight roads. What distance (correct to the nearest kilometre) is saved by going directly from Springfield to Shellbyville instead of travelling via Capital City? PS R C



8 **a** What is the value of cos 90°?

b What does $c^2 = a^2 + b^2 - 2ab \cos C$ simplify to if $C = 90°$?

c Hence what happens to the cosine rule when it is applied to a right-angled triangle?

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The cosine rule for angles

8.10

If we rewrite the cosine rule so that $\cos A$ is the subject, then we will have a formula for finding an unknown angle when the 3 sides of a triangle are known.

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$c^2 + 2ab\cos C = a^2 + b^2$$ Adding $2ab\cos C$ to both sides so that $\cos C$ appears on the LHS

$$2ab\cos C = a^2 + b^2 - c^2$$

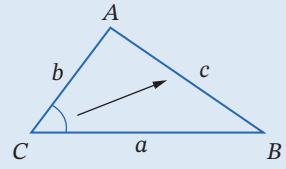
$$\therefore \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

The cosine rule for angles

For any triangle ABC:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

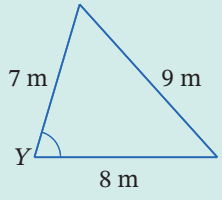
where C is the unknown angle, c is the side opposite C, and a and b are the other 2 sides.



The cosine rule can be used to find an unknown angle if the lengths of the 3 sides are known.

Example 23

Find the size of the marked angle Y, correct to the nearest degree.



Solution

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

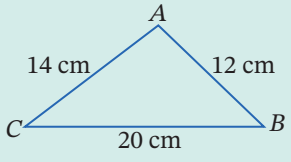
$$\cos Y = \frac{7^2 + 8^2 - 9^2}{2 \times 7 \times 8} = \frac{32}{112}$$ 9 m is opposite angle Y

$$Y = 73.398...$$

$$\approx 73°$$ From the diagram, an answer of 73° looks reasonable

Example 24

Calculate, correct to the nearest minute, the size of the largest angle in this triangle.



Solution

The largest angle is opposite the longest side, so it is $\angle A$.

$$\cos A = \frac{14^2 + 12^2 - 20^2}{2 \times 14 \times 12} = \frac{-60}{336}$$ Cos is negative so the angle will be obtuse

$$A = 100.28656...$$

$$\approx 100° 17'$$ From the diagram, an answer of 100° 17′ looks reasonable

PATH

8.10

Worksheet
Cosine rule problems

Video
The cosine rule for angles 2

Videos
The cosine rule
The cosine rule for angles 1

PATH

EXERCISE 8.10 ANSWERS ON P. 654

The cosine rule for angles U F PS R

1 Solve each equation for X, correct to the nearest degree.

a $\cos X = \dfrac{12^2 + 14^2 - 15^2}{2 \times 12 \times 14}$

b $\cos X = \dfrac{5.7^2 + 6.8^2 - 3.7^2}{2 \times 5.7 \times 6.8}$

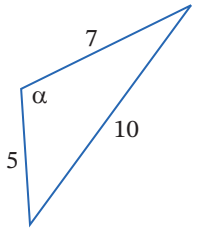
c $\cos X = \dfrac{5^2 + 6^2 - 9^2}{2 \times 5 \times 6}$

d $\cos X = \dfrac{9.2^2 + 4.7^2 - 12.8^2}{2 \times 9.2 \times 4.7}$

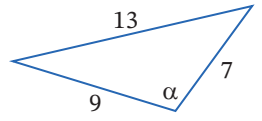
EXAMPLE 23

2 Find the size of α in each triangle, correct to the nearest degree.

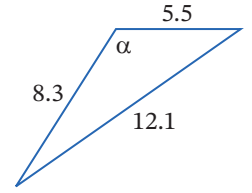
a



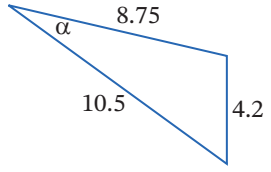
b



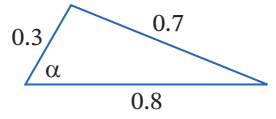
c



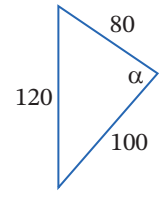
d



e



f

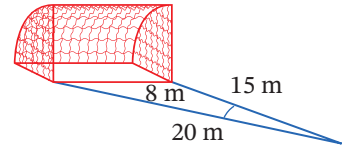


3 In $\triangle ABC$, $a = 8.5$ cm, $b = 12.1$ cm and $c = 17.6$ cm. What is the size of $\angle B$? Select the correct answer **A**, **B**, **C** or **D**.

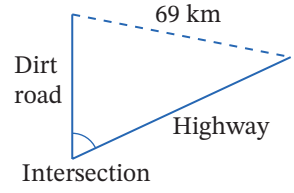
A 25° 40′ **B** 141° 57′ **C** 116° 17′ **D** 38° 3′

EXAMPLE 24

4 A soccer goal is 8 m wide. A player shoots for goal (along the ground) when 20 m from one post and 15 m from the other post. Within what angle (correct to the nearest 0.1 degree) must the shot be made for the player to have a chance of scoring a goal? PS R



5 2 cars leave an intersection at the same time. Car A drives down the dirt road at 60 km/h and car B drives down the highway at 100 km/h. After 45 minutes they are 69 km apart. Find the angle between the 2 roads, correct to the nearest minute. PS R



6 A triangle has sides of 21 m, 17 m and 10 m. Find the size of the largest angle, correct to the nearest degree. R

7 A plane flies from Alice Springs on a bearing of S65°W. After 450 km, the plane changes to a northwesterly direction for 1100 km. When the plane stops it is 1380 km from Alice Springs. What is the bearing (to the nearest degree) of the plane from Alice Springs? PS R

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The area of a triangle

8.11

PATH

8.11

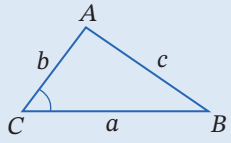
We already know that the formula for the area of a triangle is $A = \frac{1}{2}bh$, but there is also a trigonometric formula if we know the lengths of 2 sides of the triangle and the size of the **included angle** between them.

We've learned about **included angles** with the congruent and similar triangles tests.

The area of a triangle

$A = \frac{1}{2}ab\sin C$

where C is the included angle between sides a and b.



Proof:

In $\triangle ABC$, draw AX for the perpendicular height, h, of the triangle.

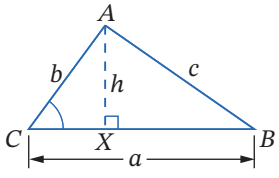
Area of $\triangle ABC = \frac{1}{2}ah$ [1]

But in $\triangle AXC$, $\sin C = \frac{h}{b}$

$\therefore h = b \sin C$

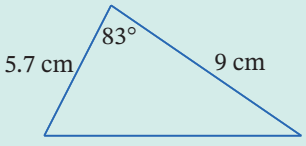
Substituting this into [1]:

$A = \frac{1}{2}ab\sin C$



Example 25

Find, correct to one decimal place, the area of this triangle.



Solution

$A = \frac{1}{2}ab\sin C$

$= \frac{1}{2} \times 5.7 \times 9 \times \sin 83°$

$= 25.458...$

$\approx 25.5 \text{ cm}^2$

83° is the included angle between 5.7 cm and 9 cm

Video
The sine area formula

PATH

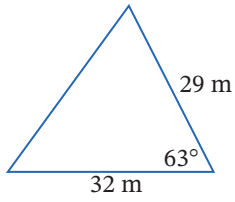
EXERCISE 8.11 ANSWERS ON P. 654

The area of a triangle U F PS R

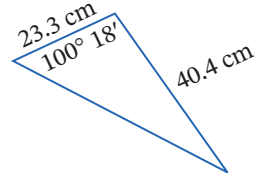
EXAMPLE 25

1 Find, correct to one decimal place, the area of each triangle.

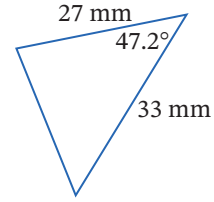
a



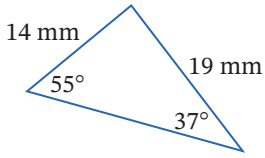
b



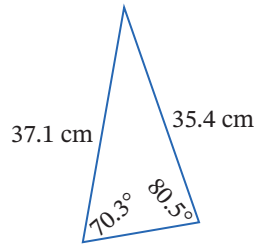
c



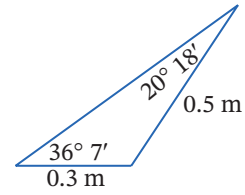
d



e

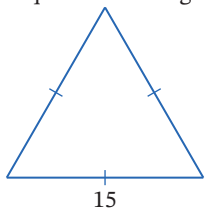


f

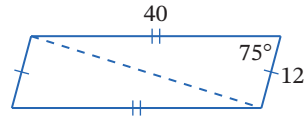


2 Calculate, correct to one decimal place, the area of each shape. All measurements are in metres. R

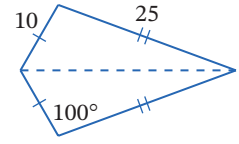
a Equilateral triangle



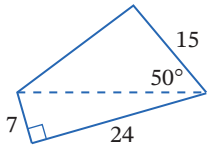
b Parallelogram



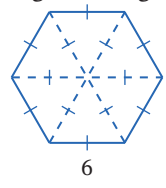
c Kite



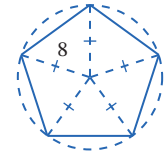
d Quadrilateral



e Regular hexagon



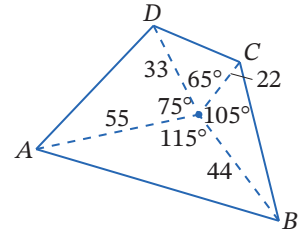
f Regular pentagon inscribed in a circle of radius 8



3 The diagram shows the results of a radial survey of a block of land. All distances are in metres. PS R

a Use the cosine rule to find the lengths of *AB*, *BC*, *CD* and *AD* and, hence, find the perimeter of the block of land (to the nearest metre).

b Use the area formula to find the area of each triangle and, hence, calculate the area of the block of land (to the nearest m²).



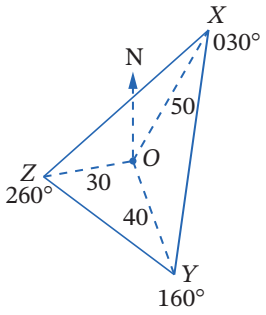
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PATH

8.11

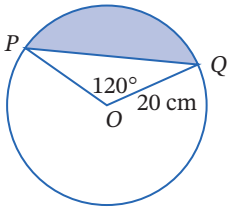
4 The results of a radial survey are shown in the diagram. All measurements are in metres. PS R



a Find the size of $\angle XOY$.

b Calculate, correct to 2 decimal places, the area of $\triangle XOY$.

5 O is the centre of a circle of radius 20 cm. Calculate, correct to one decimal place, the area of: PS R

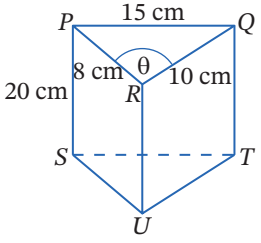


a sector OPQ

b triangle OPQ

c the shaded segment.

6 A triangular prism has base edges of 8 cm, 10 cm and 15 cm, and a height of 20 cm. PS R



a Calculate the size $\angle PRQ$, correct to nearest degree.

b Find the area of $\triangle PQR$, correct to the nearest cm^2.

c Find the volume of the prism.

7 The area of an equilateral triangle is 600 cm^2. Calculate the perimeter of the triangle, correct to one decimal place.

8 The area of ΔKLM is 24.5 cm^2. If $KL = 6.7$ cm and $KM = 8.9$ cm, find correct to the nearest degree the possible sizes of $\angle K$. PS R

9 In ΔABC, $AB = 24.1$ cm and $BC = 15.9$ cm. The area of the triangle is 115 cm^2. PS R

a Find correct to the nearest degree the possible sizes of $\angle B$.

b Find correct to one decimal place the possible lengths of AC.

Did you know?

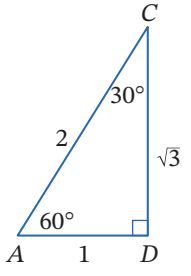
Triangle area formulas

There are many formulas for calculating the area, A, of a triangle:

Heron's formula: given 3 sides a, b, c:

$A = \sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter $\frac{a+b+c}{2}$

- Given 2 sides, a, b and the included angle, C: $A = \frac{1}{2}ab\sin C$
- Given one side, a and 3 angles, A, B, C: $A = \frac{1}{2}a^2\frac{\sin B \sin C}{\sin A}$



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1 **Find the area of this triangle using the 3 formulas, and check that your answers agree.**

2 **Show how the formula $A = \frac{1}{2}a^2 \frac{\sin B \sin C}{\sin A}$ can be proved from $A = \frac{1}{2}ab\sin C$ using the sine rule.**

3 **Research other formulas for the area of a triangle.**

8.12 Problems involving the sine and cosine rules

PATH

Video
The sine and cosine rules

Worksheets
Finding an unknown side 2

Finding an unknown angle

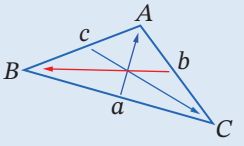
The sine and cosine rules

Puzzle
The sine and cosine rules

The sine and cosine rules

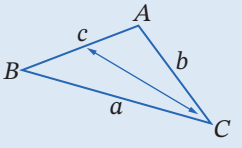
The sine rule is used for triangle problems involving 2 sides and the 2 angles opposite them.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



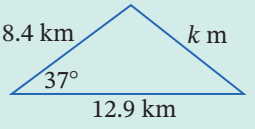
The cosine rule is used for triangle problems involving 3 sides and one angle.

$$c^2 = a^2 + b^2 - 2ab\cos C \text{ and } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

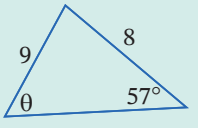


Example 26

a Find the value of k, correct to one decimal place.



b Find the value of θ, correct to the nearest minute.



Solution

a The problem involves 3 sides and one angle so use the cosine rule.

$k^2 = 8.4^2 + 12.9^2 - 2 \times 8.4 \times 12.9 \times \cos 37°$

$= 63.889...$

$k = \sqrt{63.889...}$

≈ 8.0 m

b The problem involves 2 sides and the 2 angles opposite them, so use the sine rule.

$\frac{\sin\theta}{8} = \frac{\sin 57°}{9}$

$\sin\theta = \frac{8\sin 57°}{9} = 0.7454...$

$\theta = 48.2007...$

$\approx 48° 12'$

Test whether θ could be obtuse:

$\theta = 180° - 48° 12' = 131° 48'$

But $131° 48' + 57° = 188° 48' > 180°$

So $\theta \approx 48° 12'$

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EXERCISE 8.12 ANSWERS ON P. 654

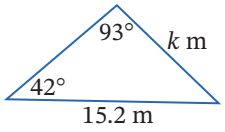
Problems involving the sine and cosine rules U F PS R C

EXAMPLE 26

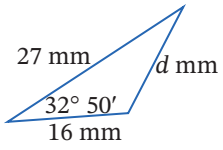
8.12

1 Find, correct to one decimal place, the value of each variable.

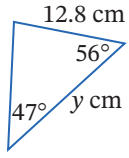
a



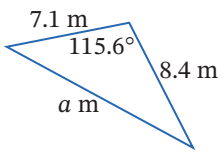
b



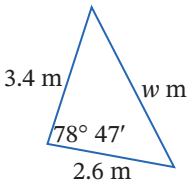
c



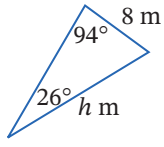
d



e

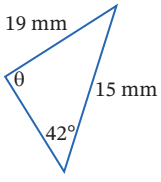


f

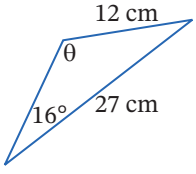


2 Find the value of θ to the nearest degree. Use diagrams to note whether θ is acute or obtuse.

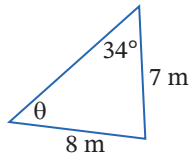
a



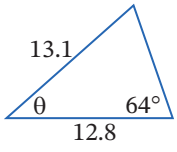
b



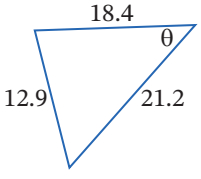
c



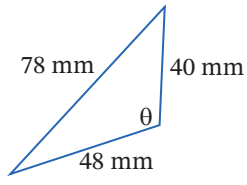
d



e



f

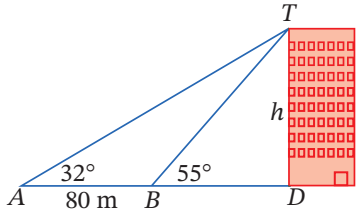


3 The angles of elevation of a building measured from 2 positions 80 m apart are 32° and 55°. PS R C

a Explain why $\angle ATB = 23°$.

b Find, correct to 2 decimal places, the length of BT.

c Hence find the height, h, of the building, correct to the nearest metre.



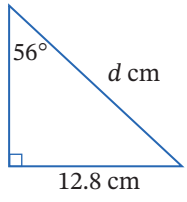
4 **a** What is the value of sin 90°? PS R C

b Find, correct to one decimal place, the value of d using:

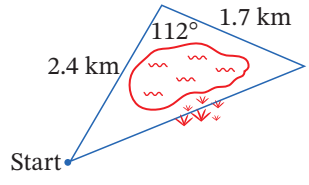
i the sine rule

ii the sine ratio for right-angled triangles.

c What do you notice about your results? Give reasons.



5 Mikayla needs to run around a cross-country course as shown. What is the length of the course, correct to one decimal place? PS R

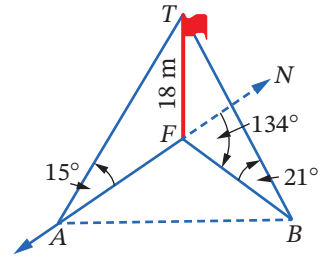


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6 A plane flew on a bearing of 150° for 370 km. It then changed direction and flew another 285 km on a bearing of 235°. How far, correct to the nearest kilometre, is the plane from its starting point? PS R

7 From a point A, south of a flagpole, the angle of elevation to the top of the pole is 15°. From B, the angle of elevation to the top of the flagpole is 21°.

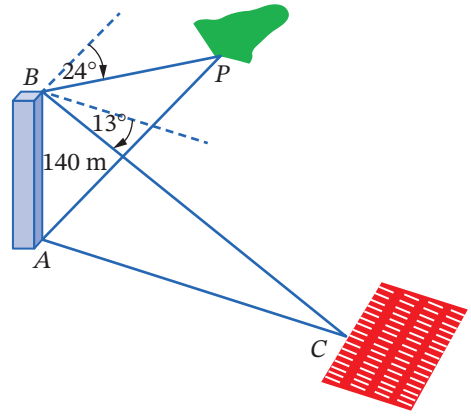


The bearing of B from the flagpole is S46°E. Find correct to one decimal place: PS R

a the length of AF and BF

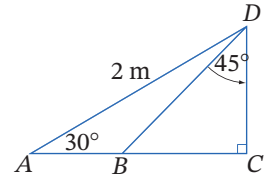
b the distance between A and B

8 From the top of a building, the angles of depression to a park (P) and a car park (C) are 24° and 13° respectively. The height of the building is 140 m, the park is due north of the building and the bearing of the car park is ESE from the building. Calculate the distance between the park and the car park, correct to the nearest metre. PS R



Power plus ANSWERS ON P. 654

1 For this diagram, find the exact value of:



a the length of AB

b the area of $\triangle ADB$

2 Find θ if θ is between 0° and 180°, and:

a $\sin\theta = \cos\theta$ **b** $\sin\theta = \sqrt{3}\cos\theta$ **c** $\sqrt{3}\sin\theta = \cos\theta$

d $\cos\theta = -\sqrt{3}\sin\theta$ **e** $\sin\theta = -\sqrt{3}\cos\theta$ **f** $\cos\theta = -\sin\theta$

g $\sin^2\theta = \frac{3}{4}$ **h** $\tan^2\theta = \frac{1}{3}$ **i** $\cos^2\theta = \frac{1}{2}$

3 Show that:

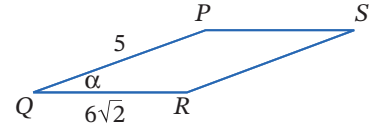
a $\sin(180° - B) + \cos(90° - B) = 2\sin B$

b $\frac{\sin(180° - B)}{\cos(180° - B)} = -\tan B$ **c** $\sin(180° - B)\cos(180° - B)\tan(180° - B) = \sin^2 B$

d $\frac{(\sin A)^2 + \sin A\cos A}{(\cos A)^2 + \sin A\cos A} = \tan A$

4 Express $\cos 45° + \sin 45°$ as a surd in simplest form.

5 The area of the parallelogram shown is 30 cm^2. Find 2 possible values for α, correct to the nearest degree.



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CHAPTER 8 REVIEW

Quiz
Language of maths 8

Puzzle
Further trigonometry crossword (Advanced)

Language of maths

angle of depression	angle of elevation	bearing	compass bearing
complementary	cosine rule	degree (°)	denominator
exact ratio	included angle	minute (′)	obtuse
Pythagoras' theorem	right-angled	second (″)	sine rule
slant height	supplementary	surd	tangent (tan)
theta (θ)	three-figure bearing	trigonometric ratio	unit circle

1 Copy and complete: A **bearing** is an _____ used to describe precisely the _____ of one location from a given reference point.

2 The word **minute** has an alternative pronunciation and meaning. What is its alternative meaning?

3 If 2 angles have a sum of 90°, what special name do we give them?

4 What is an **exact ratio**? Give an example of one.

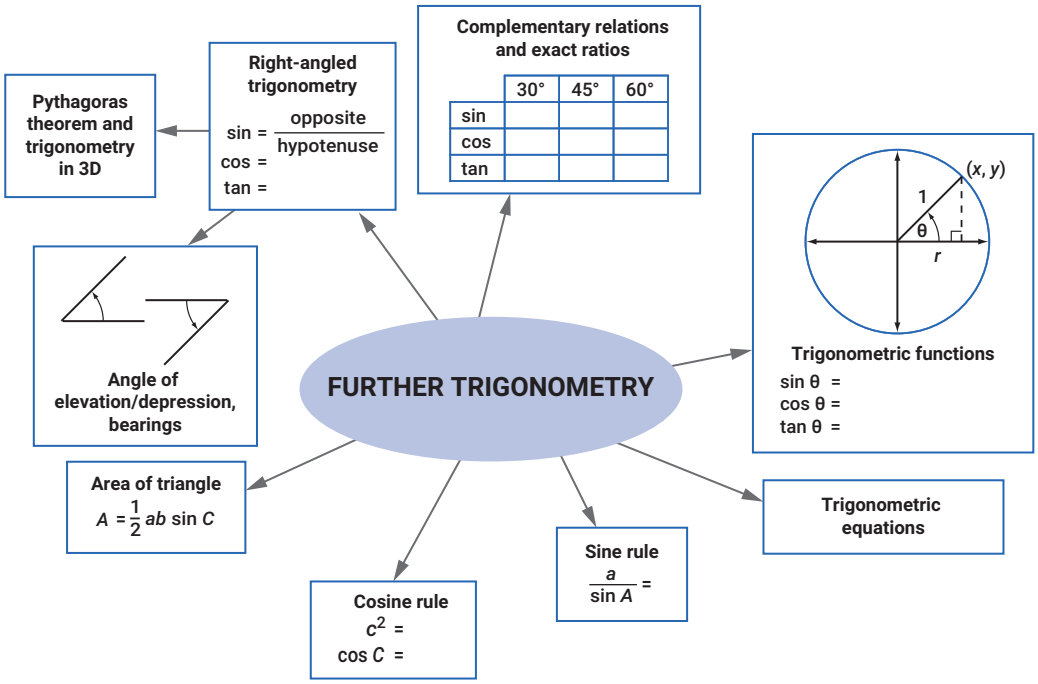
5 Describe the shape of the **sine curve**.

6 What is the name given to the angle that is between 2 known sides?

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Further trigonometry (Advanced)



TEST YOURSELF 8

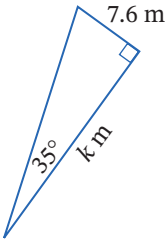
ANSWERS ON P. 655

Quiz
Test yourself 8

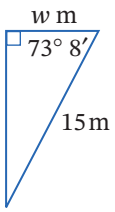
8.01

1 Find, correct to one decimal place, the value of each variable.

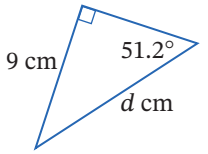
a



b



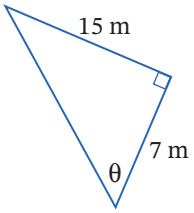
c



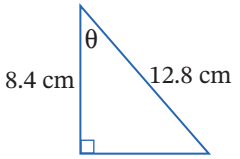
8.01

2 Find the value of θ, correct to the nearest minute.

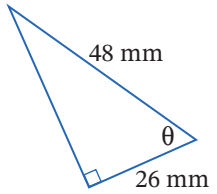
a



b



c



8.01

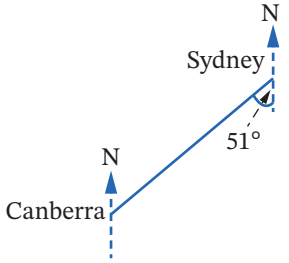
3 From the top of a 300 m cliff Renee observes a boat at a distance of 1.3 km from the base of the cliff. Find, correct to the nearest degree, the angle of depression of the boat.

8.02

4 What is the true bearing of:

a Canberra from Sydney?

b Sydney from Canberra?



8.02

5 2 planes leave an airport at the same time. The first travels on a bearing of S26°W at 450 km/h. The second travels on a bearing of N64°W at 600 km/h.

a How far apart are the planes after 3 hours, to the nearest km?

b Calculate, correct to the nearest degree, the compass bearing of the first plane from the second plane.

PATH

8.03

6 A box in the shape of a square prism has a base of 10 cm by 10 cm and is 30 cm tall. Find to the nearest whole number:

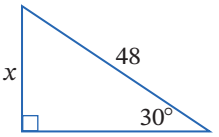
a the length of the longest diagonal of the box

b the angle that the longest diagonal makes with the base.

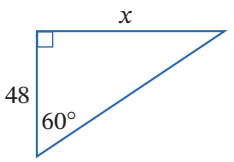
8.04

7 Find the exact value of x in each triangle.

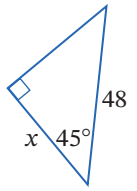
a



b



c



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PATH

8 Graph $y = \cos\theta$ for θ from 0° to 360°. 8.05

9 Find the acute angle α if: 8.05

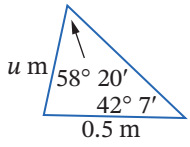
a $\sin 123° = \sin\alpha$ **b** $\tan 93° = -\tan\alpha$ **c** $\cos 110° = -\cos\alpha$

10 Solve each trigonometric equation, giving all possible acute and obtuse solutions correct to the nearest minute. 8.06

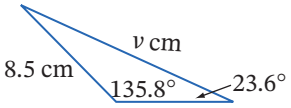
a $\cos x = -\frac{4}{5}$ **b** $\tan x = \frac{15}{7}$ **c** $\sin x = \frac{3}{14}$

11 Find the value of each variable, correct to one decimal place. 8.07

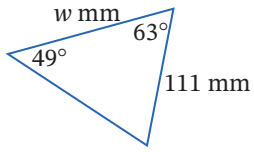
a



b

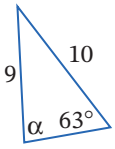


c

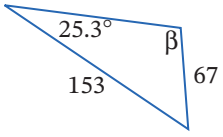


12 Find the value of each variable, correct to the nearest minute. 8.08

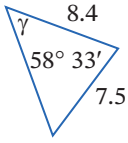
a



b

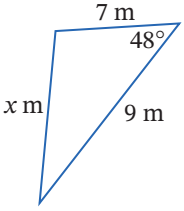


c

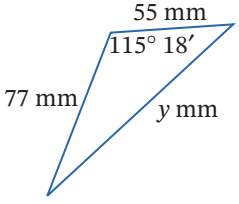


13 Find the value of each variable, correct to one decimal place. 8.09

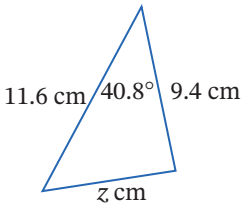
a



b

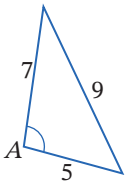


c

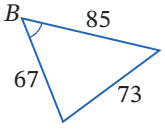


14 Find the size of each marked angle, correct to the nearest degree. 8.10

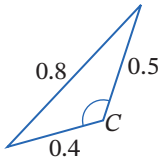
a



b

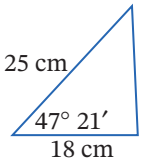


c

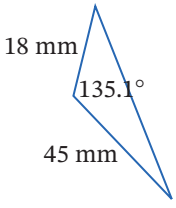


15 Find, correct to the nearest whole number, the area of each triangle. 8.11

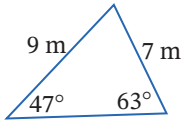
a



b

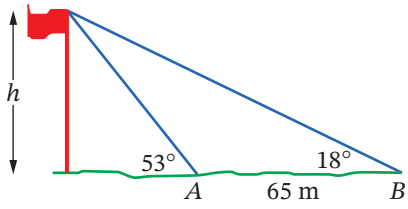


c



16 A parallelogram has sides of 12 cm and 6 cm and one interior angle of 65° 35′. Find the length of the longer diagonal, correct to one decimal place. 8.12

17 The distance AB is 65 m. Calculate the height of the flagpole, correct to one decimal place. 8.12



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